

PCS–P125 power

Rev A0 | 2026–10–06 | Schematics authored by Chinmoy Bhuyan | PCS–P125: two–level SiC bridge (6 x SG2M040170HJ per switch), split DC link, LCL, ports

01 Control connectors, defaults

PCS_PC (= PC pin for pin, 2x32) and PCS_X (2x8) to PV–CTL (PCS assembly); command pull–downs, RDY pull–up, DC– star points

File: sheets/01_pc.kicad_sch

02 DC link, bleeders, sensing

5 + 5 x C3D1U147 in two series halves, midpoint M (to the C_f star); bleeder per half; VB = V(DC+), VA = V(M) around VMID (REF3030E)

File: sheets/02_dclink.kicad_sch

03 Phase leg a, power stage

2 x 6 x SG2M040170HJ on Al2O3 pads (earthed section a); 10 k + 1 nF per gate; 3 sub–cells: 3 x 2.2 uF film + RC damper each

File: sheets/03_lega.kicad_sch

04 Phase leg b, power stage

2 x 6 x SG2M040170HJ on Al2O3 pads (earthed section b); 10 k + 1 nF per gate; 3 sub–cells: 3 x 2.2 uF film + RC damper each

File: sheets/04_legb.kicad_sch

05 Phase leg c, power stage

2 x 6 x SG2M040170HJ on Al2O3 pads (earthed section c); 10 k + 1 nF per gate; 3 sub–cells: 3 x 2.2 uF film + RC damper each

File: sheets/05_legc.kicad_sch

06 Leg a high–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/8.75 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT

File: sheets/06_gdah.kicad_sch

07 Leg a low–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/8.75 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT; bias SN6505B + T_BIAS4

File: sheets/07_gdal.kicad_sch

08 Leg b high–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/8.75 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT

File: sheets/08_gdbh.kicad_sch

09 Leg b low–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/8.75 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT; bias SN6505B + T_BIAS4

File: sheets/09_gdbl.kicad_sch

10 Leg c high–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/8.75 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT

File: sheets/10_gdch.kicad_sch

11 Leg c low–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/8.75 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT; bias SN6505B + T_BIAS4

File: sheets/11_gdcl.kicad_sch

12 Phase a LCL filter, current sensor

Switch node -> L1 (chassis) -> sensor -> C_f node F_A: C_f 2 x 25 uF + R_d–C_d to the star M (DC midpoint) -> L2 (chassis) -> AC port

File: sheets/12_filta.kicad_sch

13 Phase b LCL filter, current sensor

Switch node -> L1 (chassis) -> sensor -> C_f node F_B: C_f 2 x 25 uF + R_d–C_d to the star M (DC midpoint) -> L2 (chassis) -> AC port

File: sheets/13_filtb.kicad_sch

14 Phase c LCL filter, current sensor

Switch node -> L1 (chassis) -> sensor -> C_f node F_C: C_f 2 x 25 uF + R_d–C_d to the star M (DC midpoint) -> L2 (chassis) -> AC port

File: sheets/14_filtc.kicad_sch

15 AC voltage sensing

6 x (6 x 1 M ARHV06 + 4.99 k to VMID, 4.7 nF): VC1–3 at the C_f nodes, VG1–3 at the AC terminal taps; OPA2388 followers to PCS_X

File: sheets/15_acsense.kicad_sch

16 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/16_leanB_power.kicad_sch

17 Port B sensing and interlocks

Created BUS–master and slave domains, polarity and dV interlocks, domains: LIVE (DC– = GND = AGND, the AC side included), GD_<leg><H/L> gate islands (functional: NSI6651, T_BIAS4), PE (pads, heatsink, NTC lugs, Y / varistor / IMD crossings), SELV (AUX–T1). Stages 1–7 drawn (gen/pcs_power.py header); OPEN items listed there and in the design check.

File: sheets/17_leanB_ctrl.kicad_sch

18 Port B coil drives

Created PCS + PWR design, gate islands, passive bank bleeder, islands (functional: NSI6651, T_BIAS4), PE (pads, heatsink, NTC lugs, Y / varistor / IMD crossings), SELV (AUX–T1). Stages 1–7 drawn (gen/pcs_power.py header); OPEN items listed there and in the design check.

File: sheets/18_leanB_drive.kicad_sch

19 DC terminals, insulation test

DC terminals 250 A time, Y capacitors, board entries, removable PE link, port status to PCS_PC, insulation monitor (port.lean_imd)

File: sheets/19_dcterm.kicad_sch

20 AC port: contactors, SPD, RCM

L2 out -> K1 -> K2 -> CM choke -> AC terminals; coil drivers (fail–safe high side); 3 x TVT25751 + 3 x Y1 to PE_T; type–B RCM sensor

File: sheets/20_acport.kicad_sch

21 PE bonds, heatsink, NTC inputs

Board PE bond, heatsink sections (earthed), heatsink NTC lug probes, L1 winding NTC leads, NTC filters (biased on the control board)

File: sheets/21_earthbond.kicad_sch

22 Live 5 V / 3.3 V, SELV connector

+24V (aux live) -> +5V and +5V_GD (LMR38020), +3V3 (TPS62130 from +5V_GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board

File: sheets/22_supplies.kicad_sch

23 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = DC–; 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 2 x 3.3 uF / 1300 V film = surge absorber (AUX–HV rev C1 input block)

File: sheets/23_aux75_input.kicad_sch

24 AUX75 start–up, brown–in, OV, timer

As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 39–100 ms

File: sheets/24_aux75_start.kicad_sch

Not bench–validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()

Sheet 1

Schematics authored by Chinmoy Bhuyan

Live board referenced to DC–; gate islands functional; HV–PE basic

Sheet: 1

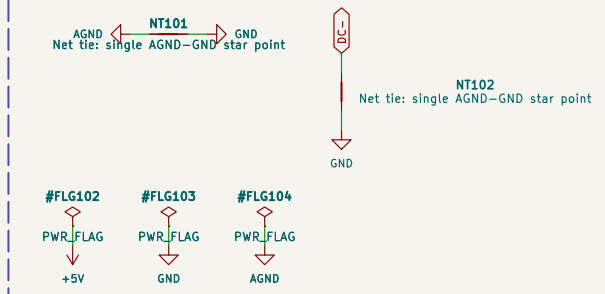
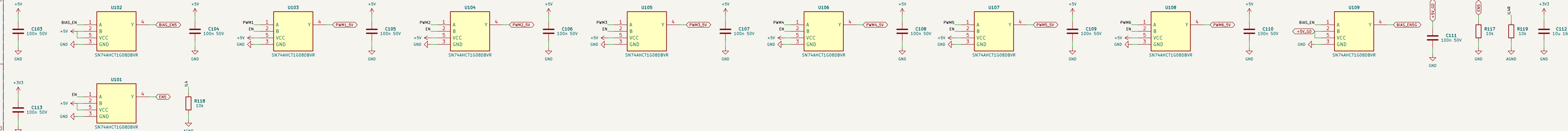
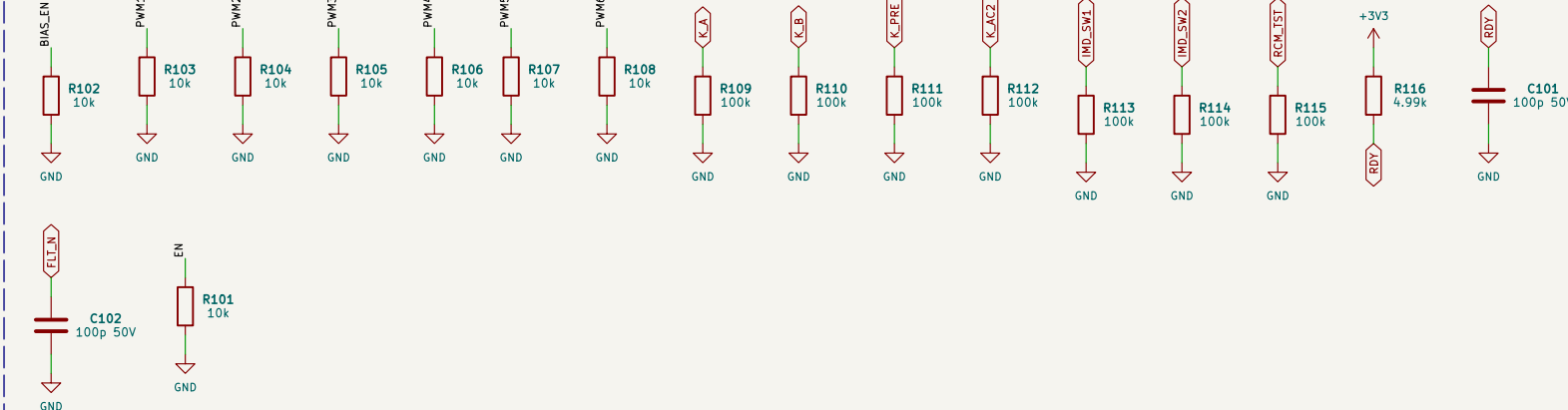
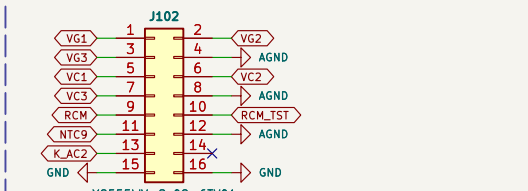
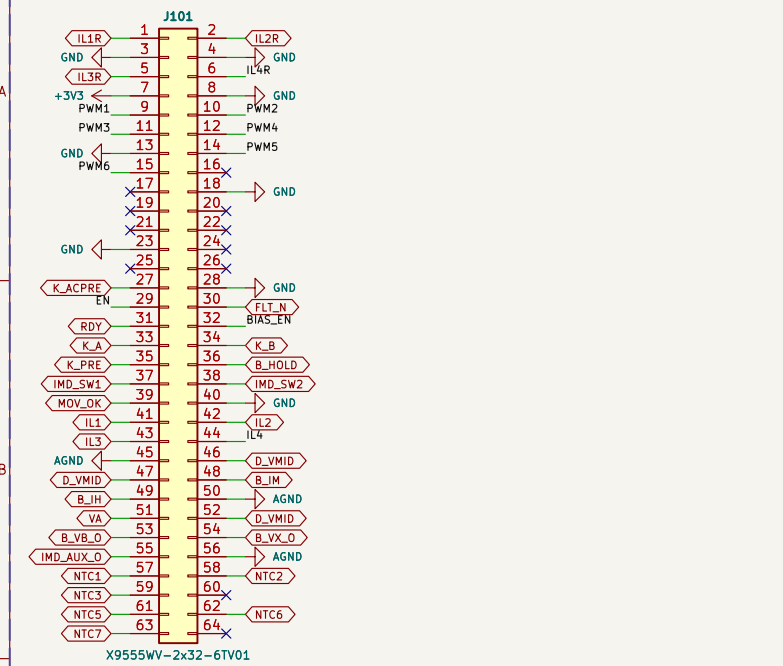
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Title

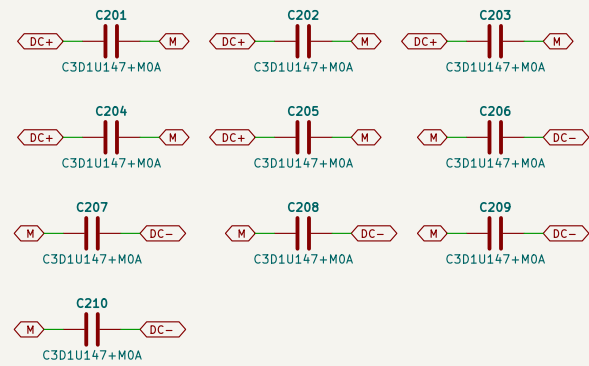
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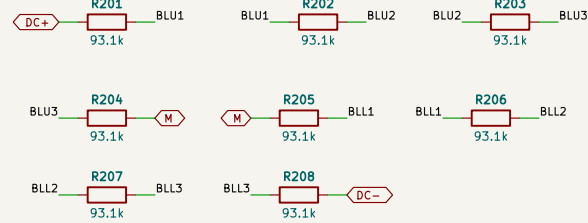


SPLIT DC LINK



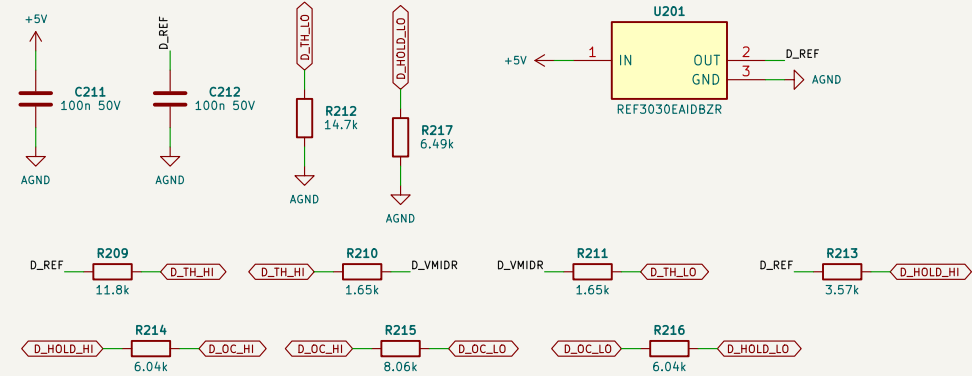
Upper half DC+ to M, lower half M to DC-, 5 x 140 uF each (700 uF per half, 350 uF in series).
M is the C_f star point of the LCL filter (no midpoint control: two-level).

BLEEDERS (BALANCE THE HALVES, DISCHARGE THE LINK)



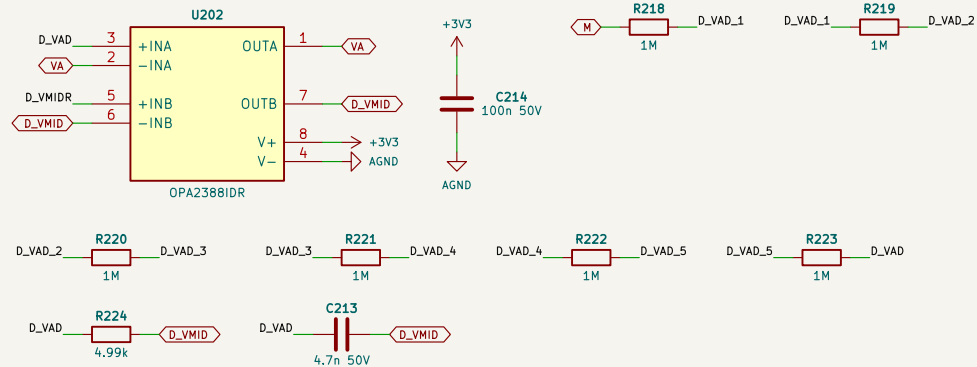
4 x 93.1k 1210 per half (372.4k); label 'wait 15 min'.

REFERENCE (PORT.LEAN_REFS: REF3030E, VMID 1.646 V AND THE DC-PORT THRESHOLD LADDERS)



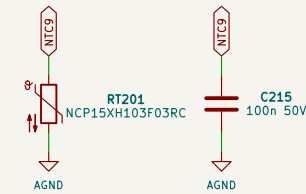
Shared with the DC port (stage 4). VMID buffered here; IA, IA_H, VAX of PCS_PC sit on it.

MIDPOINT DIVIDER (6 X 1 M ARHV06 + 4.99 K TO VMID, 4.7 NF: 1 V = 1203 V)



VA = VMID + V(M)/1203 (OPA2388 follower); its second half is the VMID follower. VB (DC+) comes from the DC port's bank divider (lean_dividers).

DC-LINK TEMPERATURE (NTC9 TO PCS_X)



Murata NCP15 10 k on the board between the film capacitors; biased on the control board.

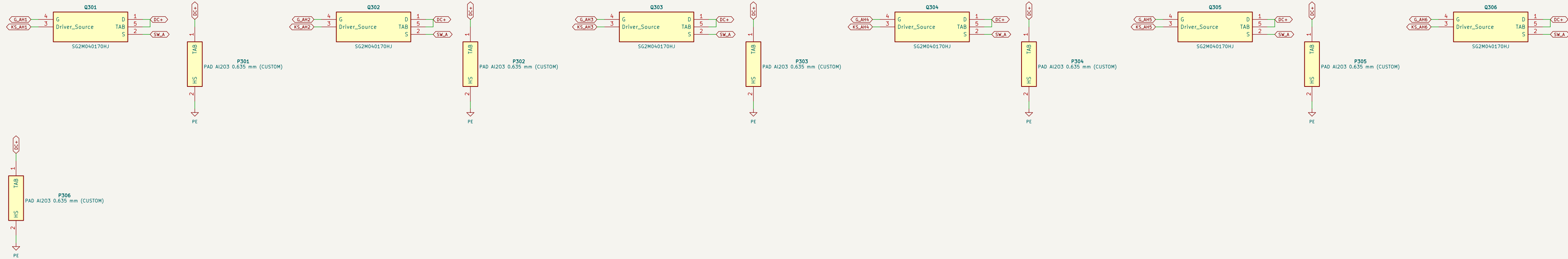
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 3
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /02 DC link, bleeders, sensing/
File: 02_dclink.kicad_sch

Title: PCS-P125 power – DC link, bleeders, sensing

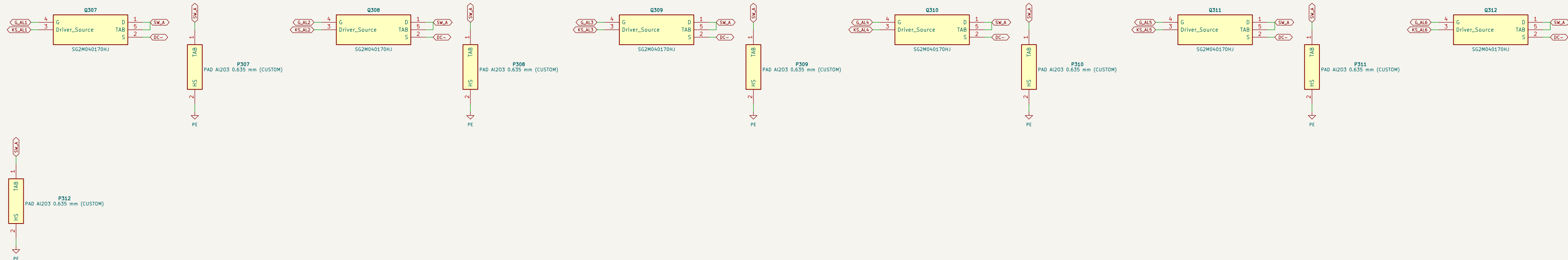
Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 3/28

LEG A HIGH SWITCH: 6 X SG2M040170HJ



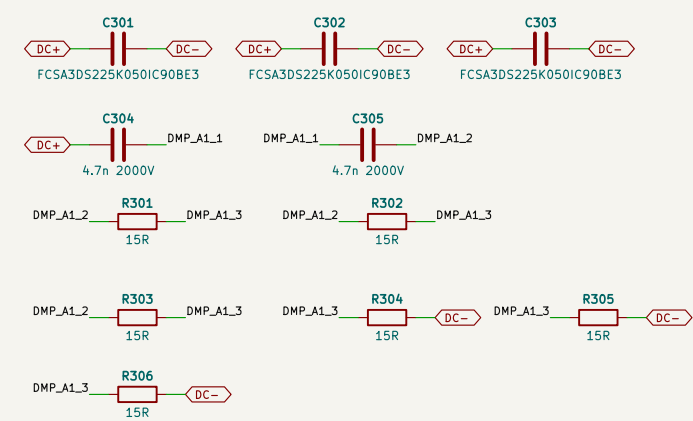
Per device: own gate net G_AH<k> and Kelvin net KS_AH<k> (gate network on the gate-drive sheet). BOM NOTE: R_DS(on) of the six within +/- 5 % (V_GS 18 V, 38 A, 25 C), V_GS(th) within +/- 0.25 V, one lot per switch (pcs_spec device_acceptance, PCM-06).

LEG A LOW SWITCH: 6 X SG2M040170HJ



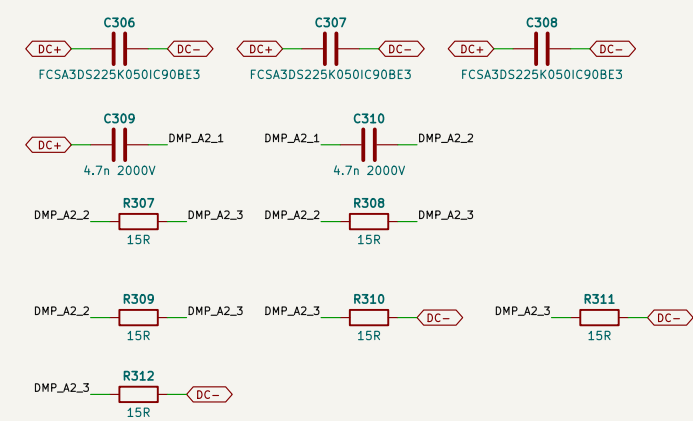
Per device: own gate net G_AL<k> and Kelvin net KS_AL<k> (gate network on the gate-drive sheet). BOM NOTE: R_DS(on) of the six within $\pm 5\%$ (V_{GS} 18 V, 38 A, 25 C), $V_{GS(th)}$ within ± 0.25 V, one lot per switch (pcs_spec device_acceptance, PCM-06).

LEG A SUB-CELL 1: DECOUPLING + RC DAMPER



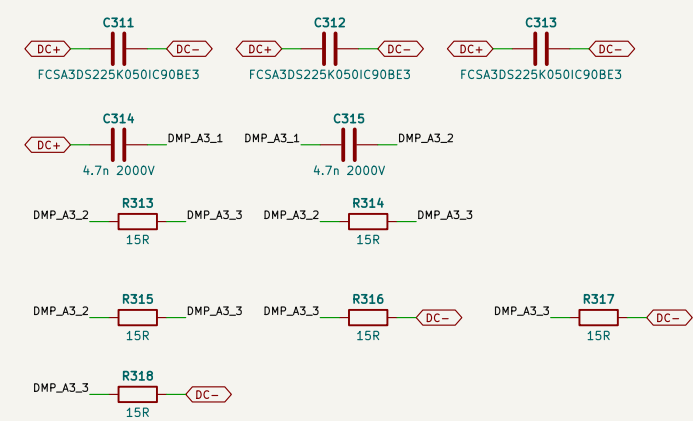
3 x FCSA3DS225 at devices 1-2 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG A SUB-CELL 2: DECOUPLING + RC DAMPER



3 x FCSA3DS225 at devices 3-4 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG A SUB-CELL 3: DECOUPLING + RC DAMPER



3 x FCSA3DS225 at devices 5-6 (high and low): damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series loop at the pins ≤ 3 nH (LAYOUT).

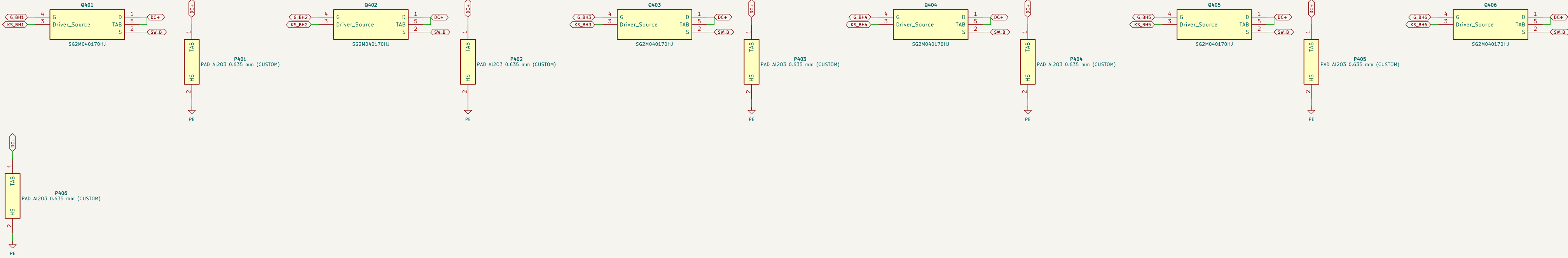
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check() Sheet 4 Schematics authored by Chinmoy Bhuyan Live board referenced to DC-; gate islands functional; HV-PE basic	H
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Sheet: /03 Phase leg a, power stage/
File: 03_lega.kicad_sch

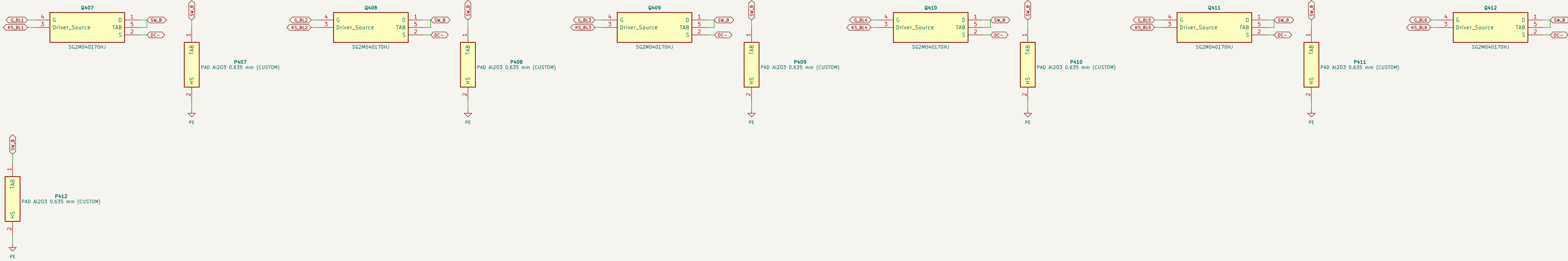
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Size: A2	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 4/28

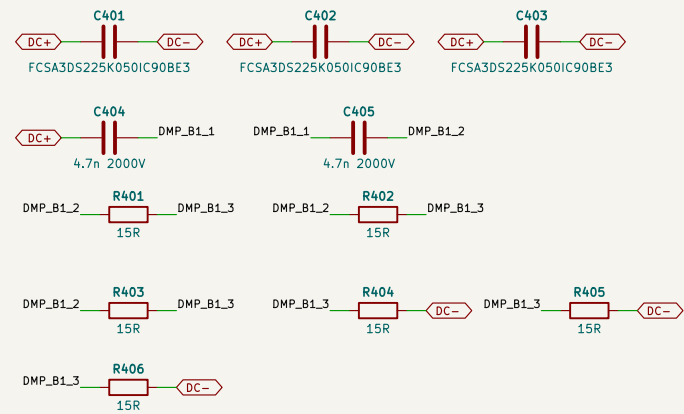
LEG B HIGH SWITCH: 6 X SG2M040170HJ



LEG B LOW SWITCH: 6 X SG2M040170HJ

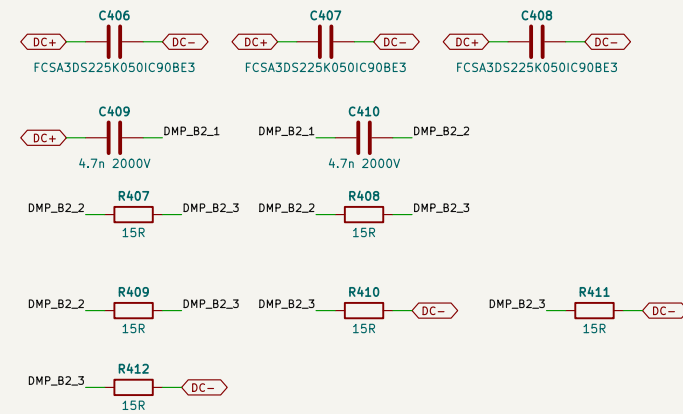


LEG B SUB-CELL 1: DECOUPLING + RC DAMPER



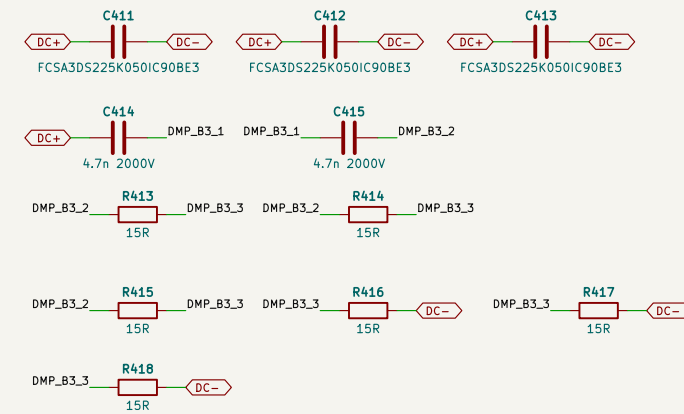
3 x FC5A30S225 at devices 1-2 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG B SUB-CELL 2: DECOUPLING + RC DAMPER



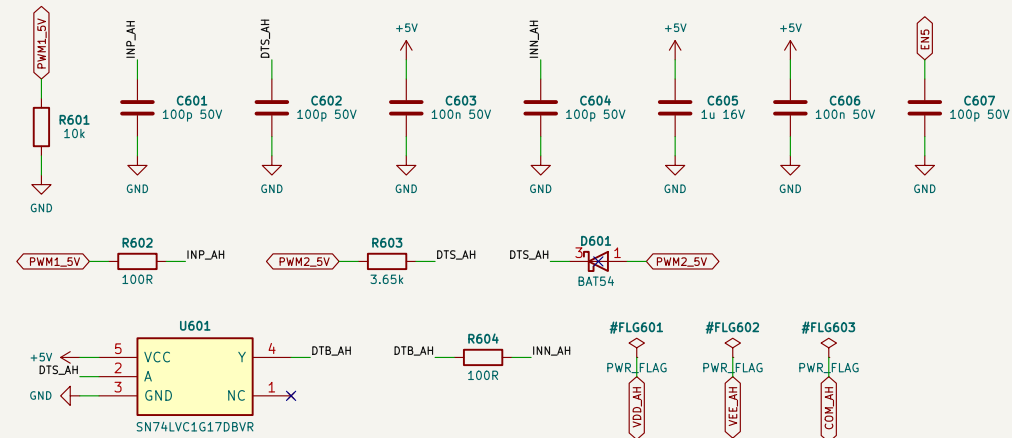
3 x FC5A30S225 at devices 3-4 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG B SUB-CELL 3: DECOUPLING + RC DAMPER



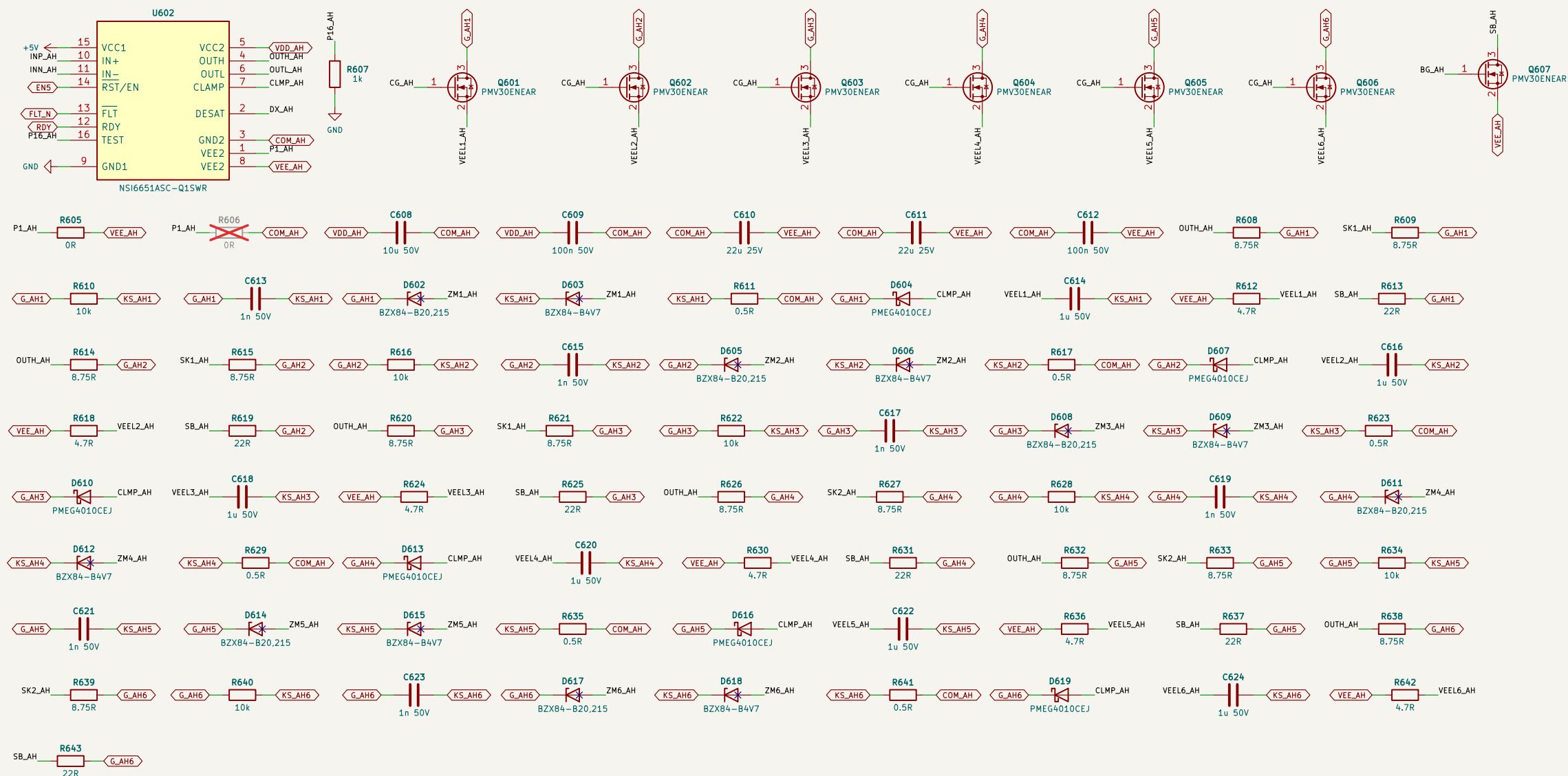
3 x FC5A30S225 at devices 5-6 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

AH: LOGIC SIDE



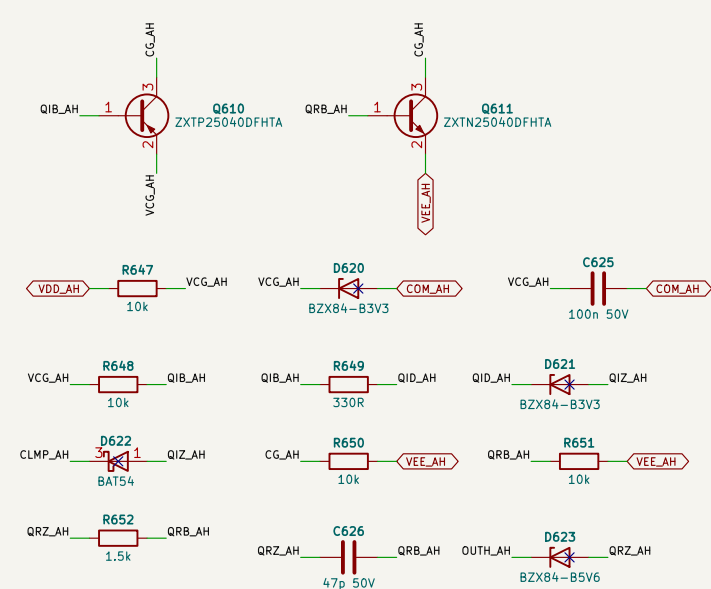
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

AH: DRIVER AND 6 GATES



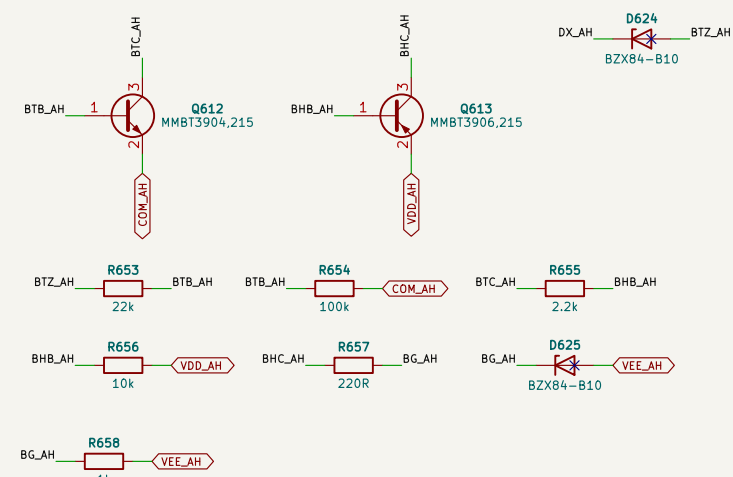
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

AH: MILLER CLAMP CONTROL



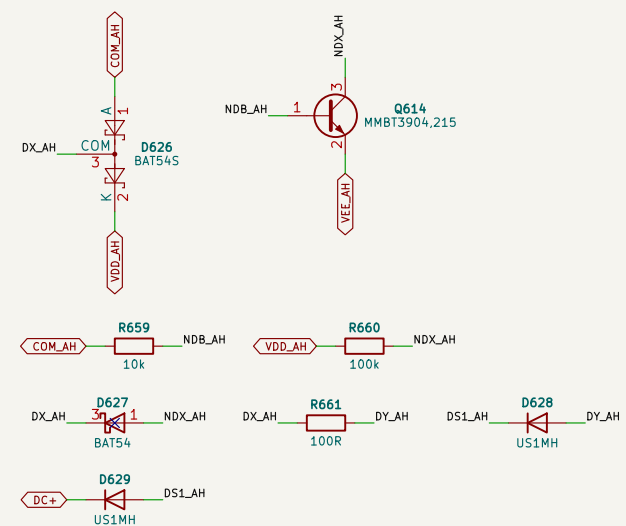
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V.
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

AH: SHORT-CIRCUIT BOOSTER



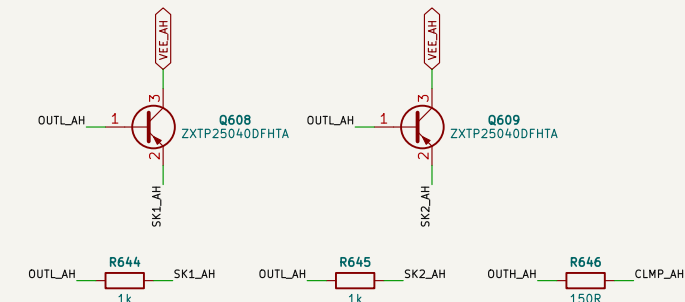
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

AH: DESAT

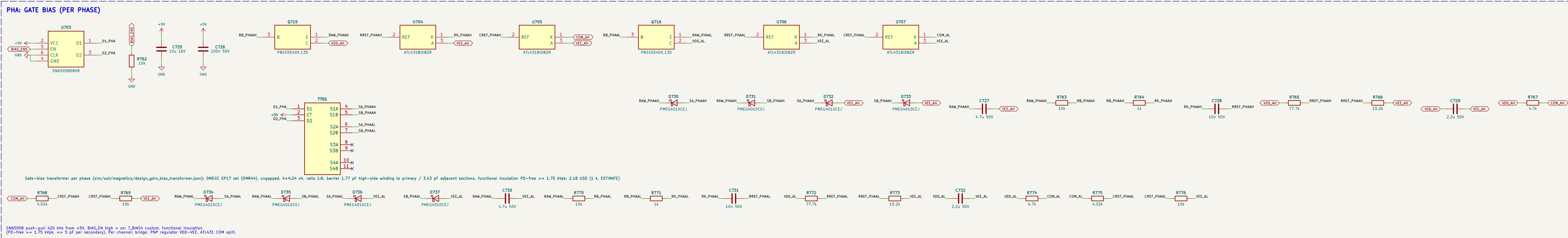
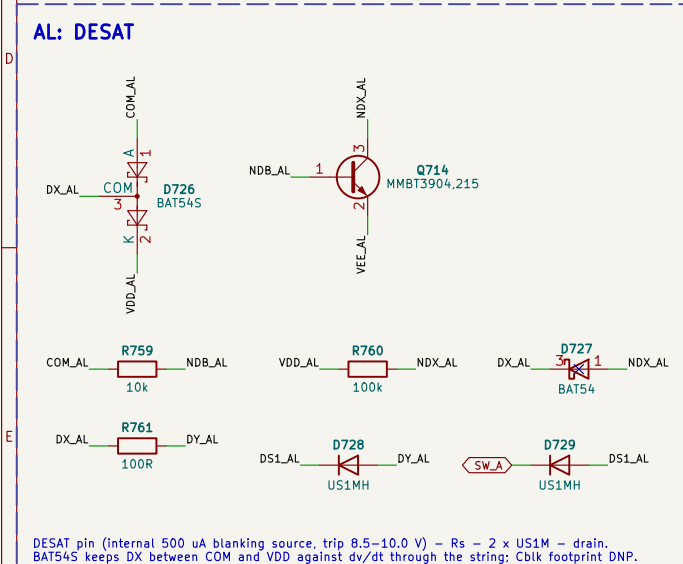
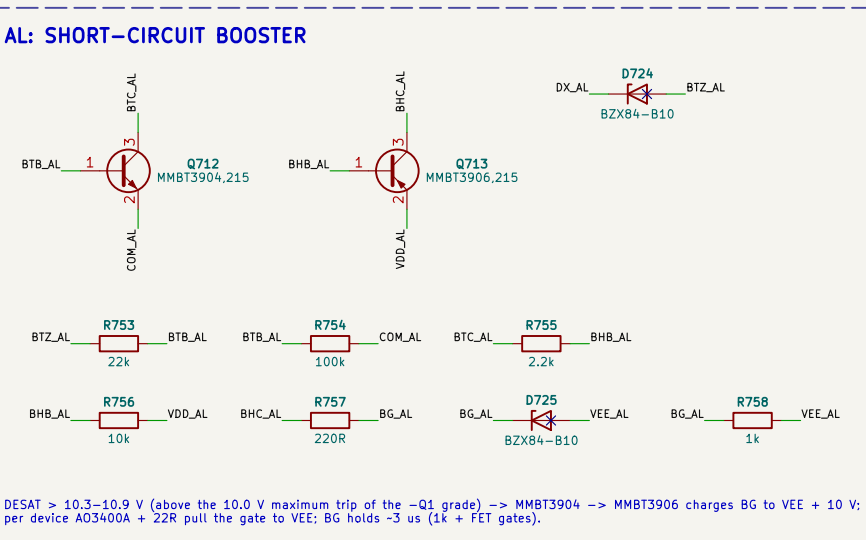
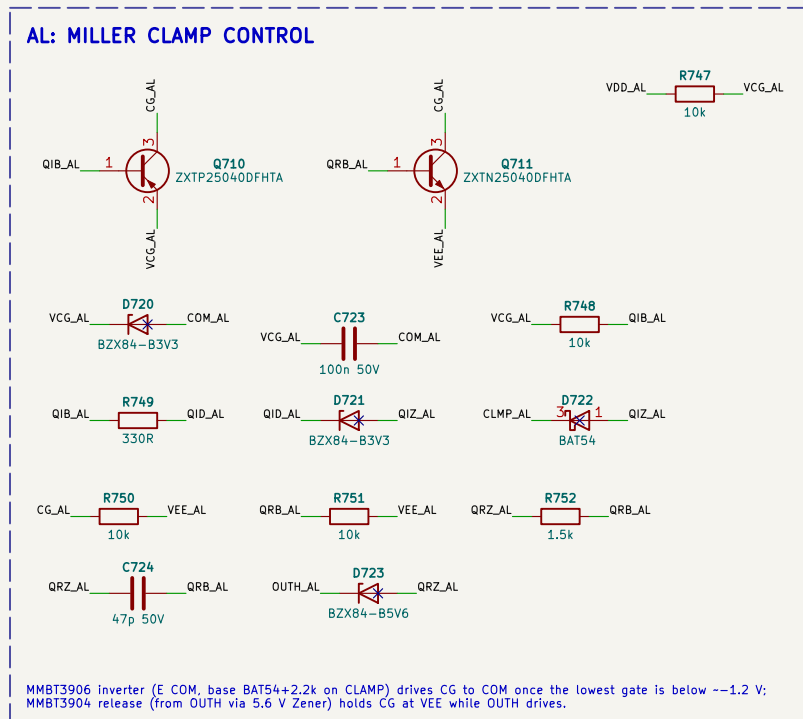
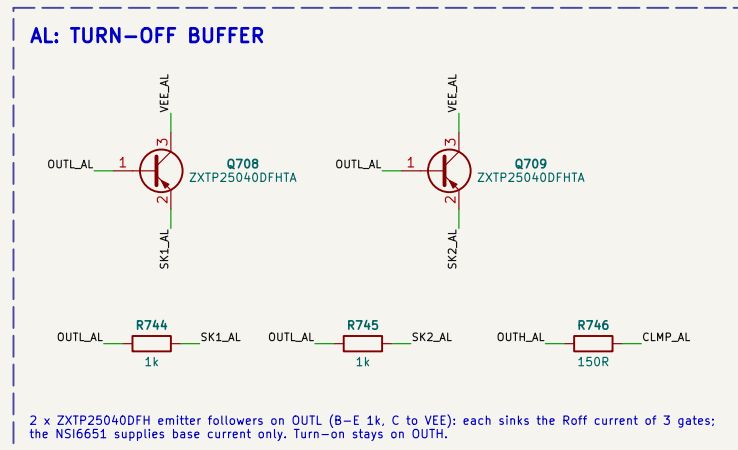
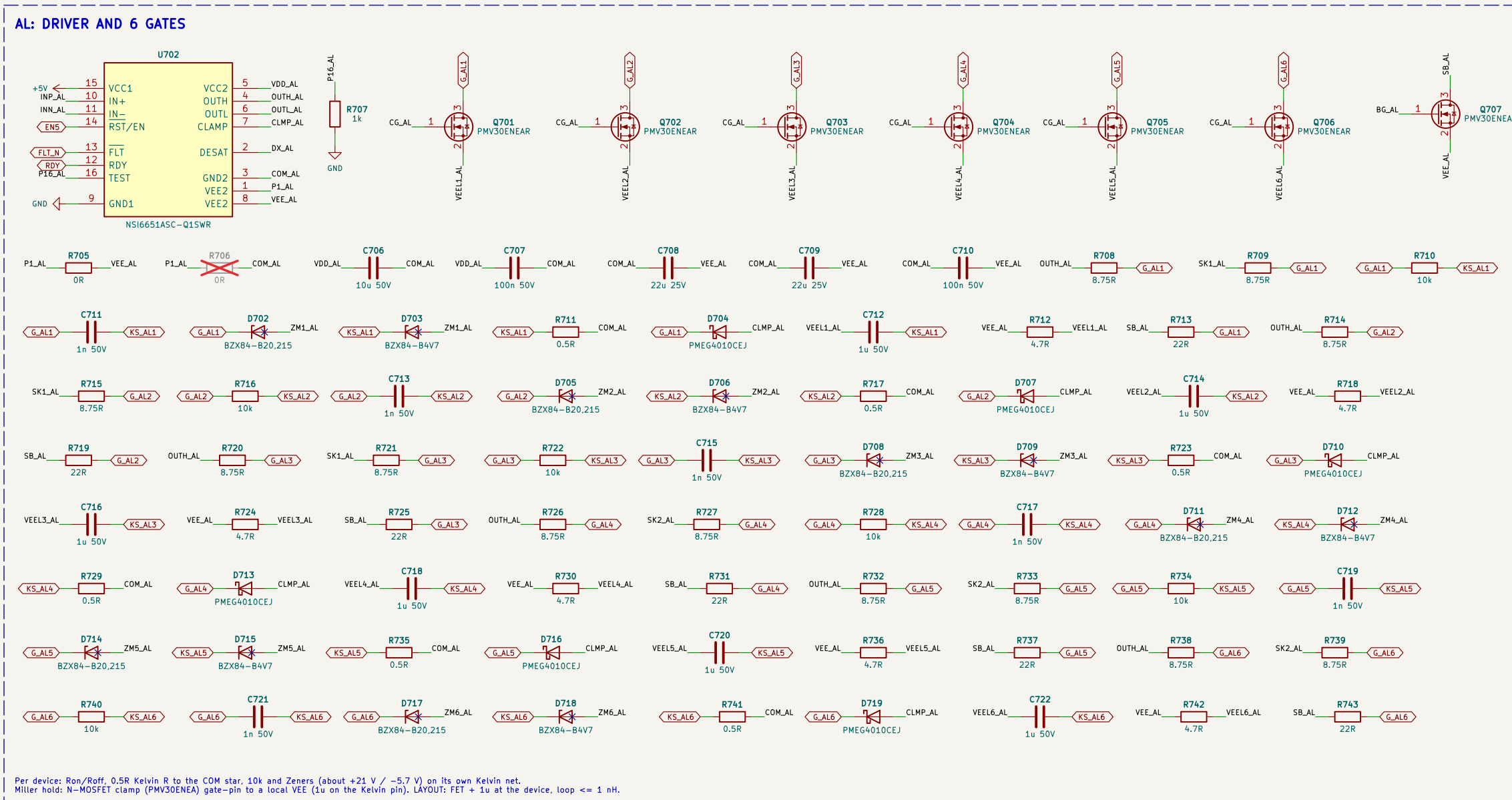
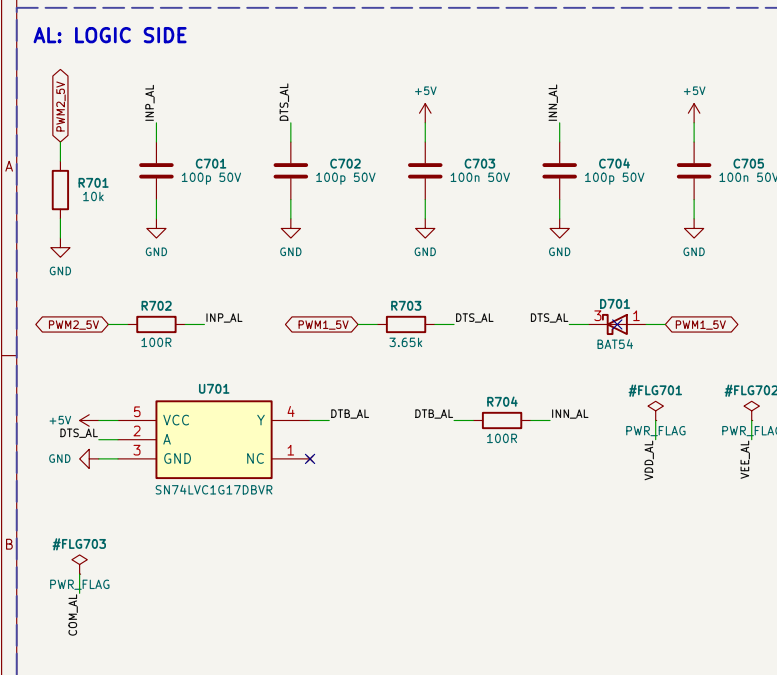


DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string: Cblk footprint DNP.

AH: TURN-OFF BUFFER



2 x ZXTPT25040DFH emitter followers on OUTL (B-E 1k, C to VEE): each sinks the Roff current of 3 gates
the NS16651 supplies base current only. Turn-on stays on OUTH.



LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length from the COM star; Ron, Roff, Kelvin R, 10k + 1n Zeners and the clamp FET + C_{CL} at each device (loop ≤ 1 nH). PNP followers next to their three gates, DESA

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_checks

Sheet: /07 Leg a low-side gate drive/
File: 07_nch_lowside.gds

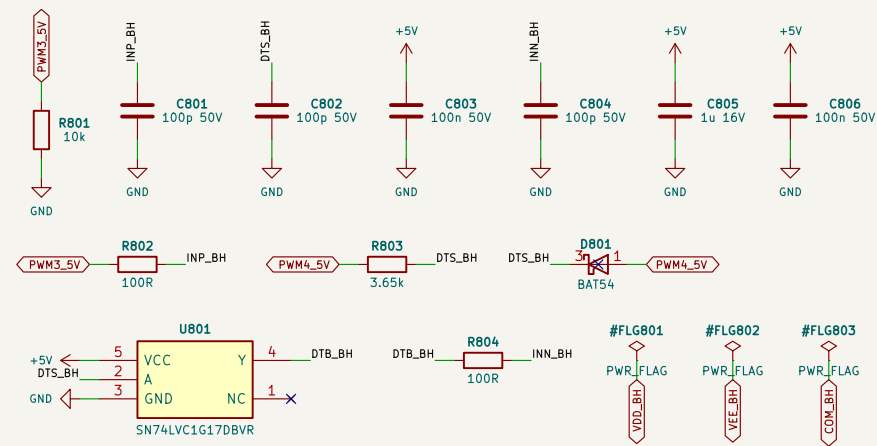
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Size: A1	Date: 2026-10-06
KiCad E.D.A. 10.0.6	

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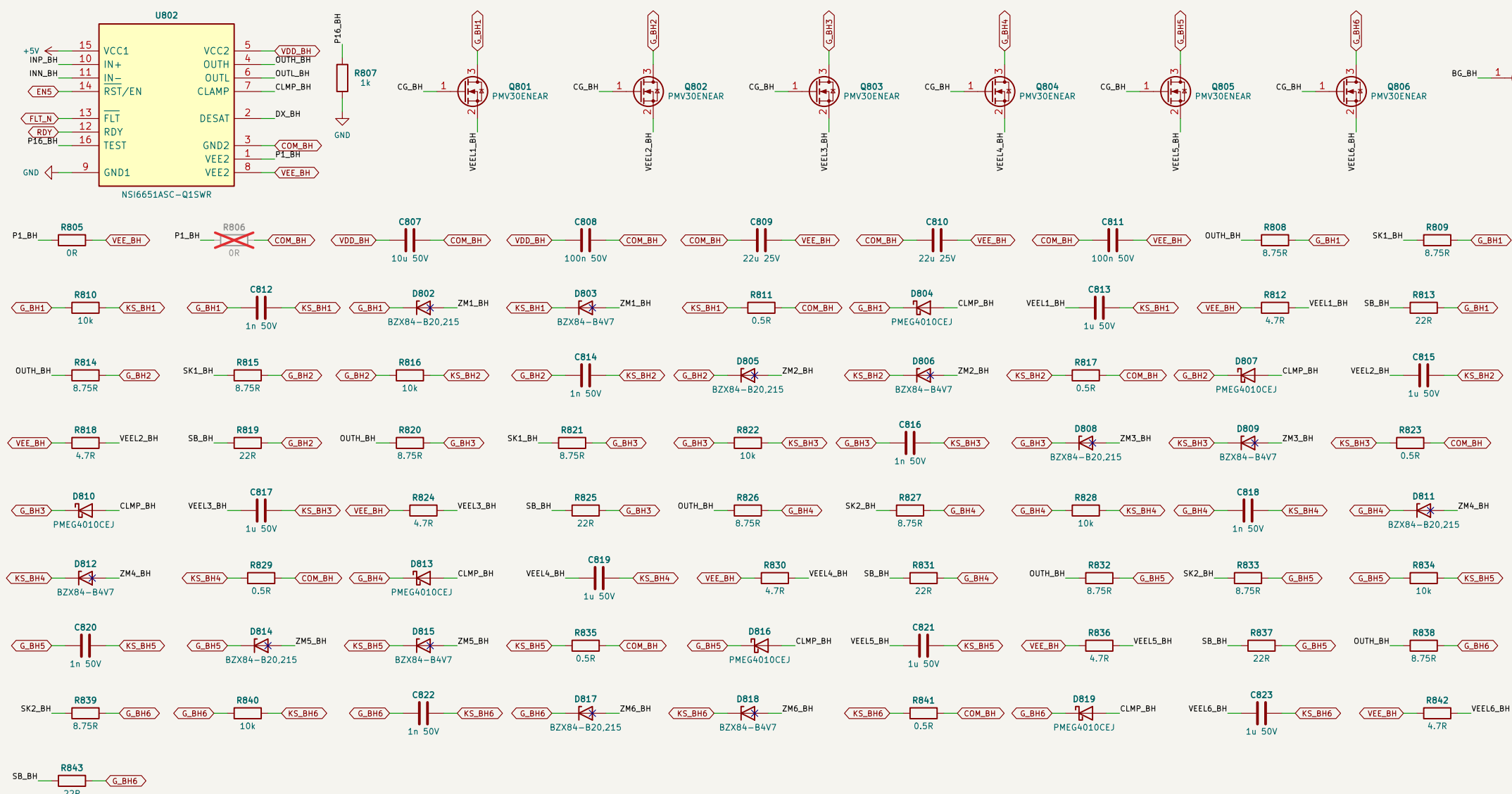
Rev: A0
Id: B/2

BH: LOGIC SIDE



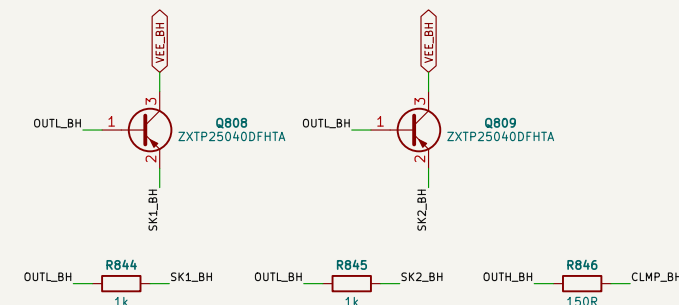
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BH: DRIVER AND 6 GATES



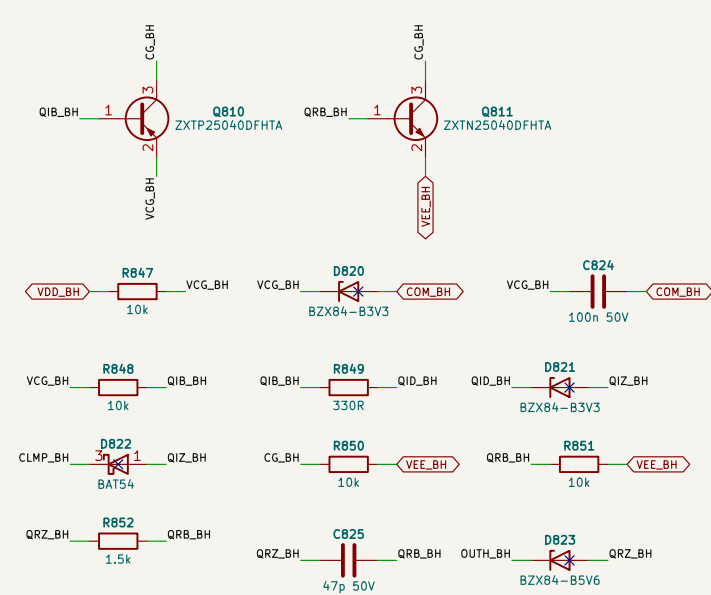
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BH: TURN-OFF BUFFER



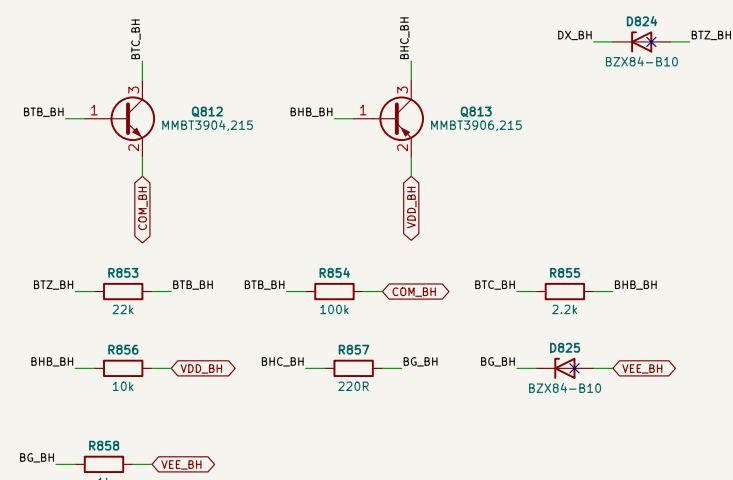
2 x XZTP25040DFH emitter followers on OUTL (B-E 1k, C to VEE): each sinks the Roff current of 3 gates; the NSI6651 supplies base current only. Turn-on stays on OUTH.

BH: MILLER CLAMP CONTROL



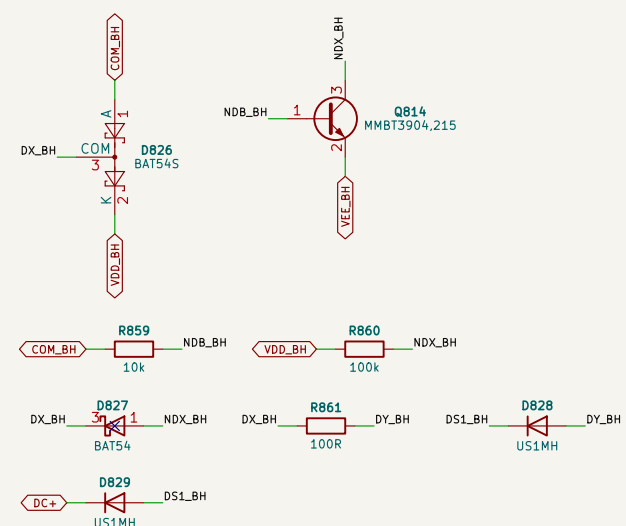
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BH: SHORT-CIRCUIT BOOSTER



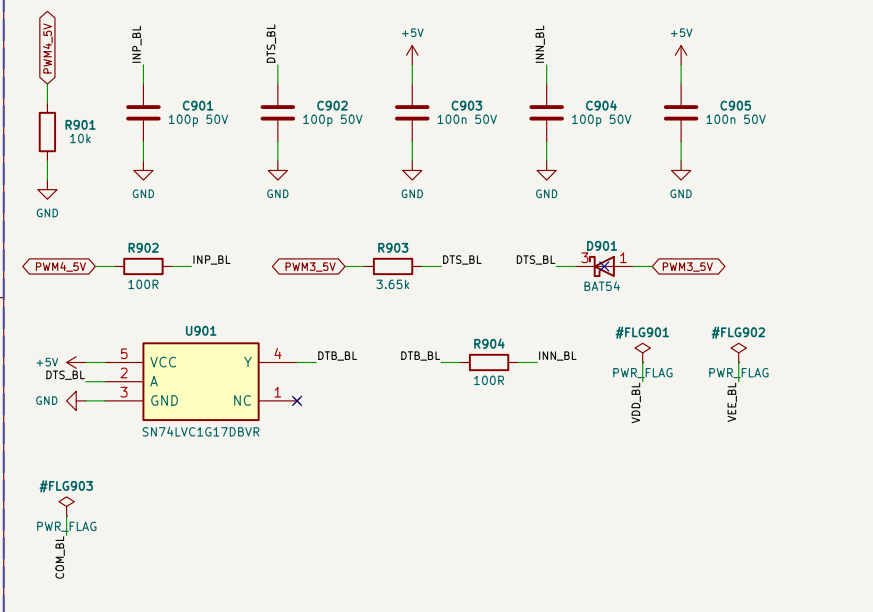
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

BH: DESAT



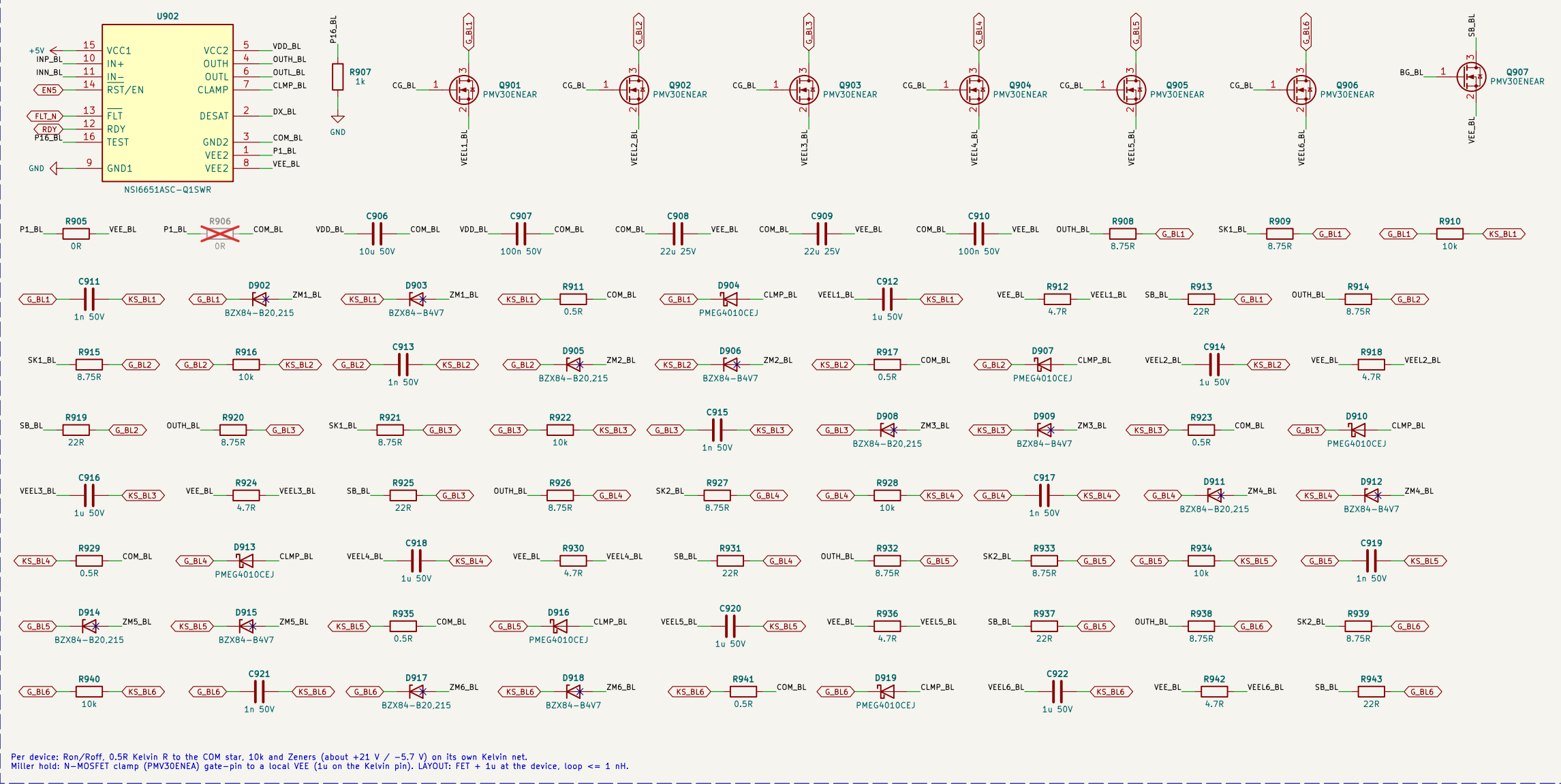
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string: Cblk footprint DNP.

BL: LOGIC SIDE



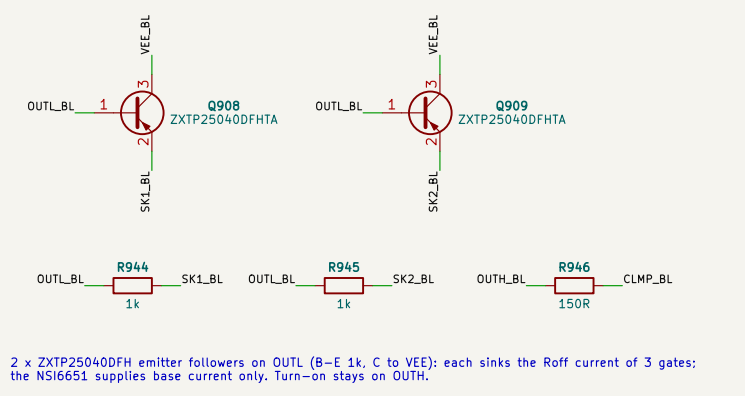
(It+ = own PWM; It- = complementary switch, falling edge stretched 3.60s/100p; C901C905 = 100pF dead time = 100pF/100pF; Data PC (active low) pulls Rdy low.

BL: DRIVER AND 6 GATES



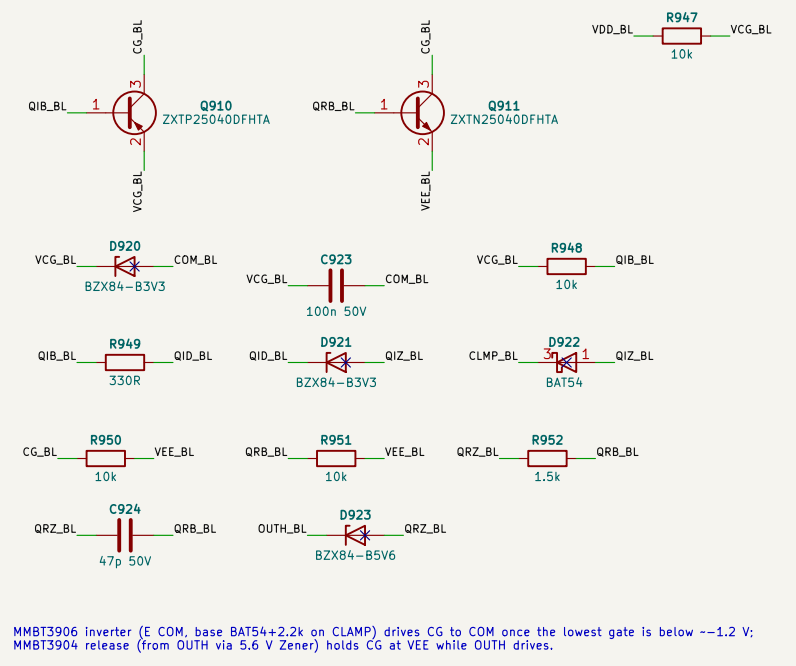
Per device: Raw/BufF: 0.58 Kilo Ohm to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kilo Ohm net. Miller hold to MOSFET clamp (PMEG4010CE) gate-pin to a base VEE (to on the Kilo Ohm pin) LATCH: 10 = 1s at the device, loop <= 1 ms.

BL: TURN-OFF BUFFER



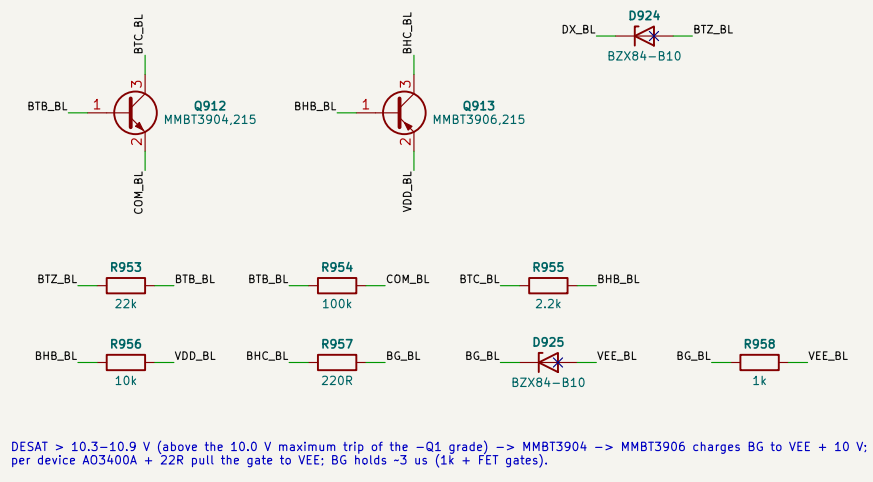
2 x ZXTF25040DPH1A another follows on OUTL (B-E 1k, C to VEE) each sinks the Roff current of 3 gates; the MOSFET supplies base current only, turn-on delay on OUTL.

BL: MILLER CLAMP CONTROL



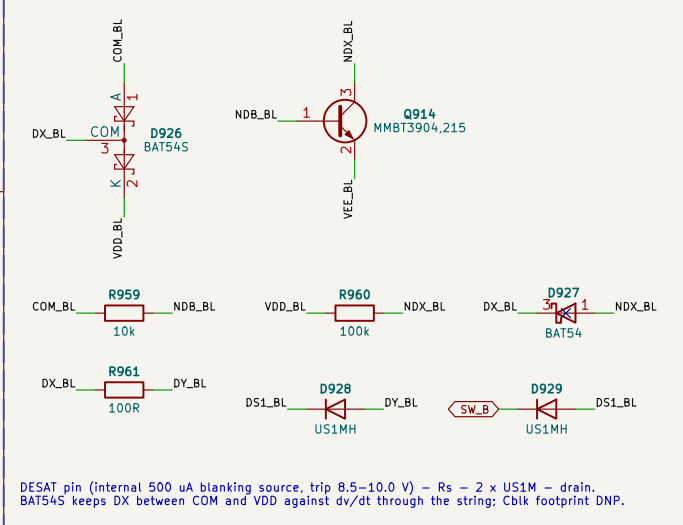
MHBT3906: Inverter (C COM, base B8204-2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V; MHBT3906: release (from OUTP via 5.6 V Zener) holds CG at VEE while OUTP driver.

BL: SHORT-CIRCUIT BOOSTER



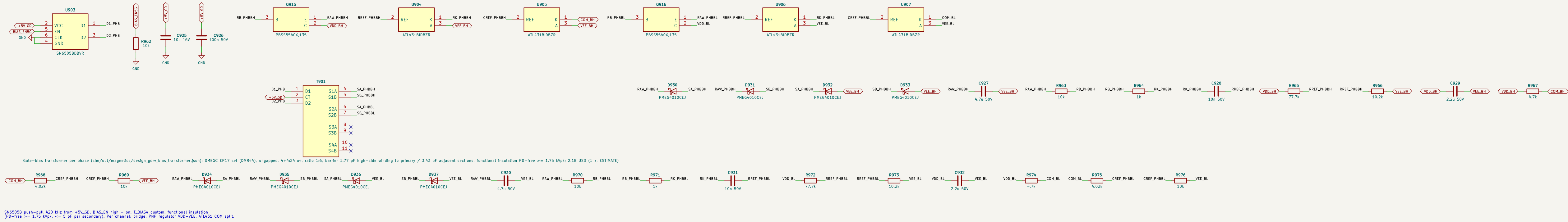
DESAT = 10.3-10.9 V (above the 10.0 V maximum tip of the -Q1 grade) -> MHBT3904 -> MHBT3906 charges BG to VEE + 10 V per device A0340A + 22M pull the gate to VEE; BG holds -3 on (1k + PFT gates).

BL: DESAT



DESAT pin (Interval 500 uA blanking source, trip 8.5-10.0 V) - Rg = 2 x USM - drain. B8204 keeps BG between COM and VEE against Rv/G through the wiring. CSB Redundant ONP.

PHB: GATE BIAS (PER PHASE)



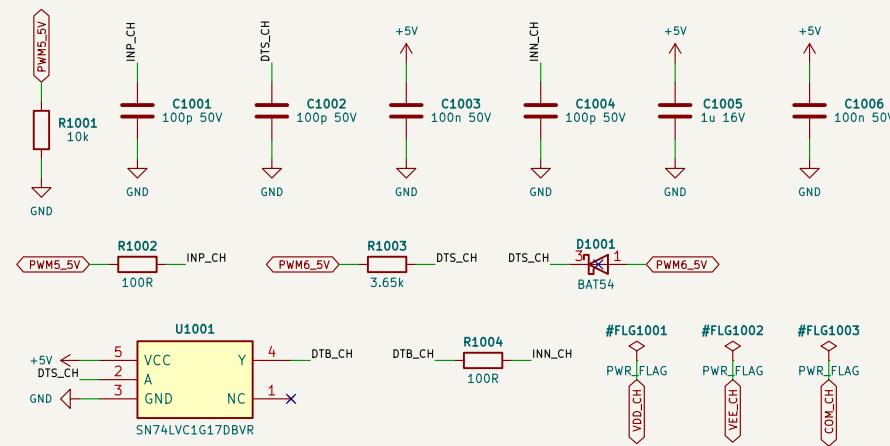
Gate-bias transformer per phase (sim/out/magnetics/design.gdn/bias.transformer.json): DMOS: EP17 set (DMR14), ungapped, 4+424 vA, ratio 1:6, barrier 1.77 pf of high-side winding to primary / 3.43 pf adjacent sections, functional insulation PD-free >= 1.75 kVp; 2.18 USU (1 k, ESTIMATE)

PHB09B push-pull 420 kVdc from -VVE,OD, B8204 high = on 1.80A custom, functional insulation (PD-free >= 1.75 kVp, <= 5 pf per secondary). Per channel bridge PNP regulatory VDD-VVE, ATLAS1 COM split.

LEG B: LAYOUT RULES

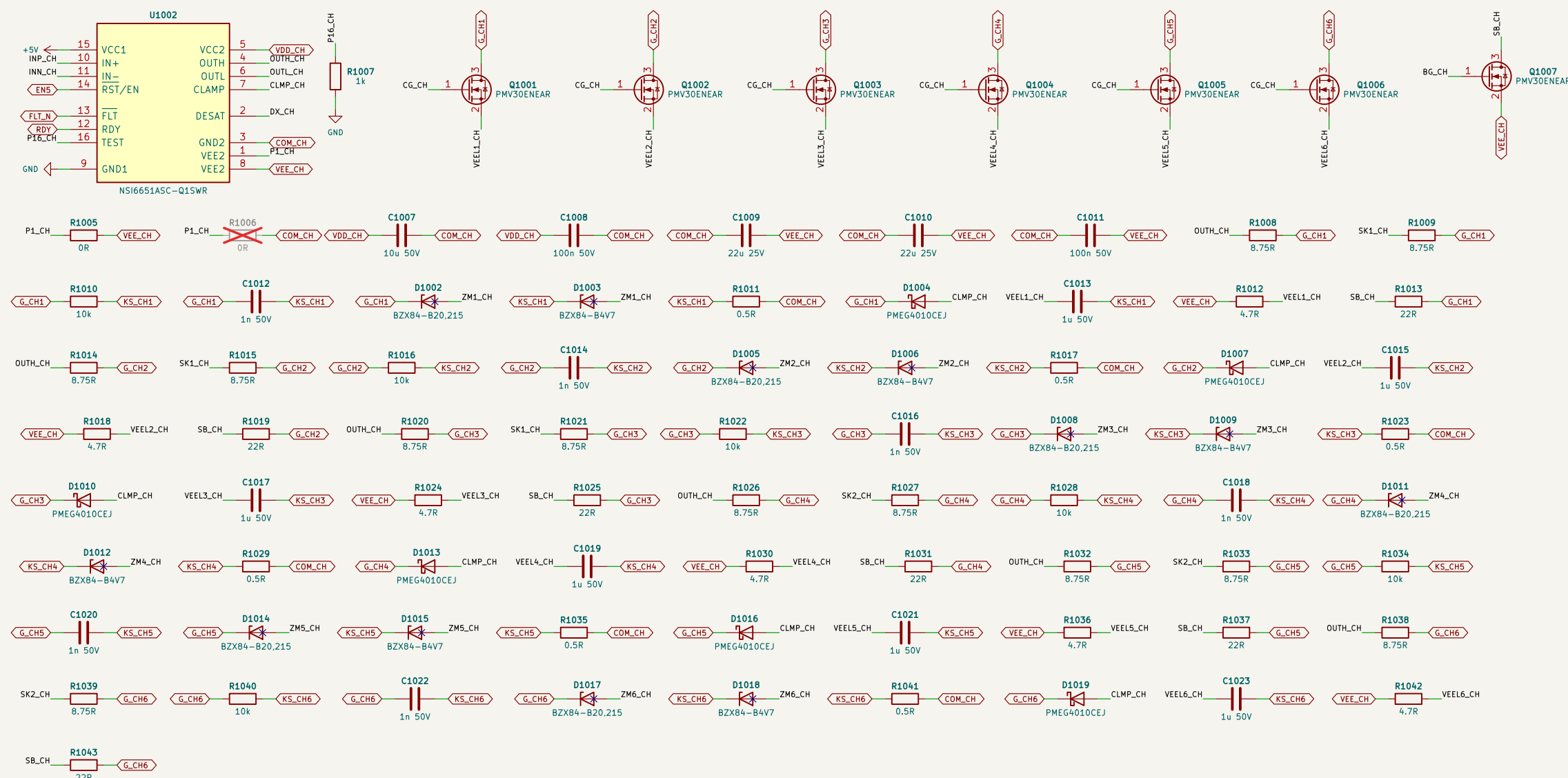
Gate and Kilo Ohm of each device as coupled pairs of equal length from the COM star; Ron: Roff: Kilo Ohm, 10k + 1n. Zeners and the clamp PFT = 2.2k, at each device (loop <= 3 m), PNP followers next to their three gates. DESAT wiring >= 2.5 mm pad gap per diode; coat to PDS. Miller hold of six gates NOT simulated - BENCH TEST.

CH: LOGIC SIDE



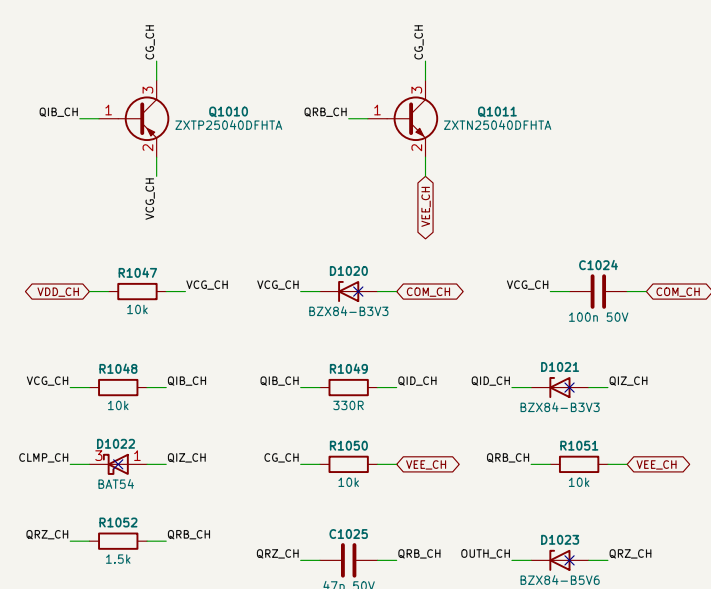
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

CH: DRIVER AND 6 GATES



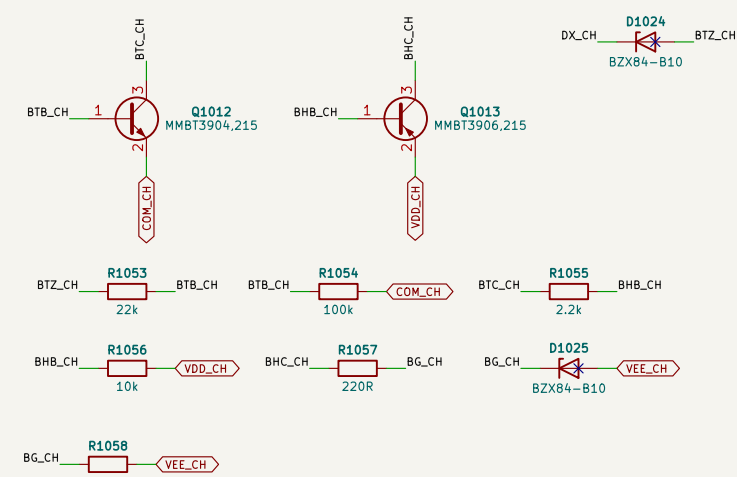
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

CH: MILLER CLAMP CONTROL



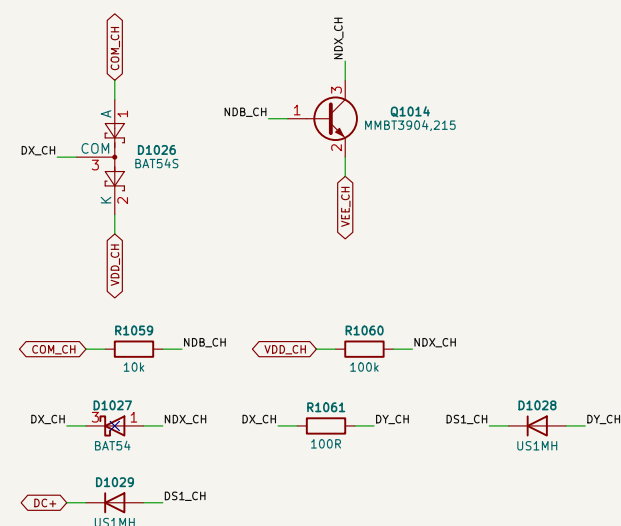
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V.
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

CH: SHORT-CIRCUIT BOOSTER



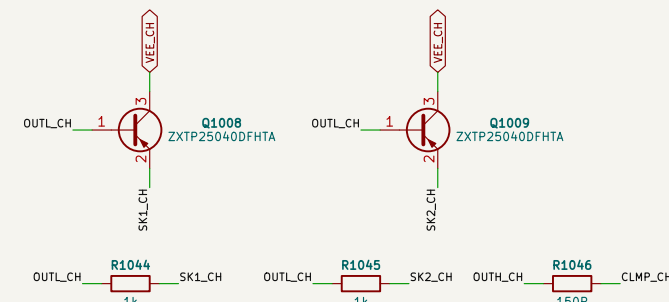
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AG3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

CH: DESAT



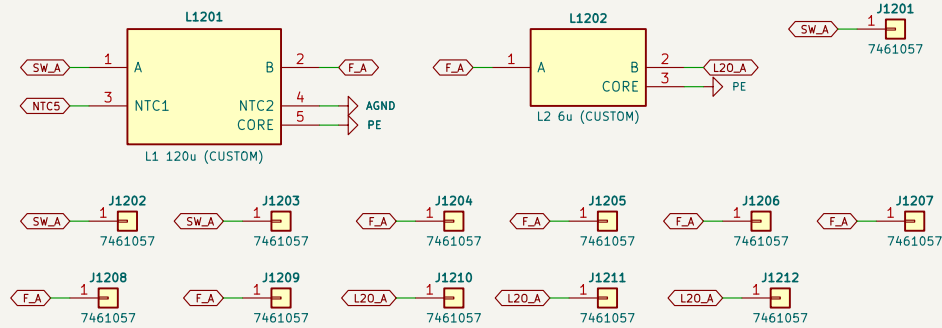
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

CH: TURN-OFF BUFFER



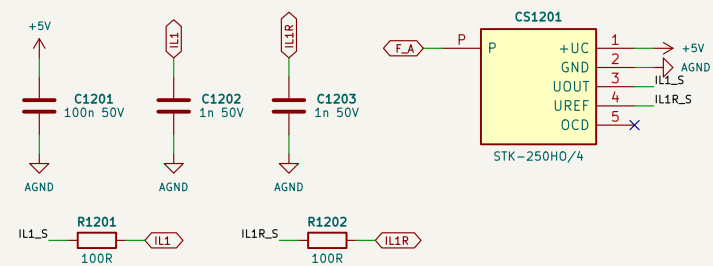
2 x ZXTPT25040DFH emitter followers on OUTL (B-E 1k, C to VEE): each sinks the Roff current of 3 gates; the NS16651 supplies base current only. Turn-on stays on OUTH.

PHASE A: L1 AND L2 (CHASSIS, CUSTOM) AND THEIR STUDS



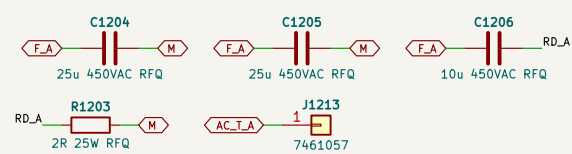
Board studs (REDCUBE press-fit, 100 A each): 3 x switch node to L1, 3 x L1 out (C_f node, its busbar through the sensor), 3 x C_f node to L2, 3 x L2 out (to AC contactor K1 on the port sheet). Cores bonded to PE.

PHASE A CURRENT SENSOR STK-250H0/4 (C_F SIDE OF L1)



IL1 = Uref + 3.2 mV/A x I (I + = L1 -> C_f; body arrow, verify at Incoming inspection). IL1R = its Uref pin: 100R + 1 nF each at the source (as PV-PWR). Primary = L1's output busbar through the aperture (a 50 Hz-smooth node, not the switch node). OCD (open drain, 2.93 x I_PN = 732 A +/-10 %) not used: above both IL windows and the L1 limit.

PHASE A C_F AND DAMPING BRANCH (STAR = M, THE DC MIDPOINT)



C_f = 2 x 25 uF MKP (50 uF); R_d 2 ohm + C_d 10 uF in series, F_A to M.

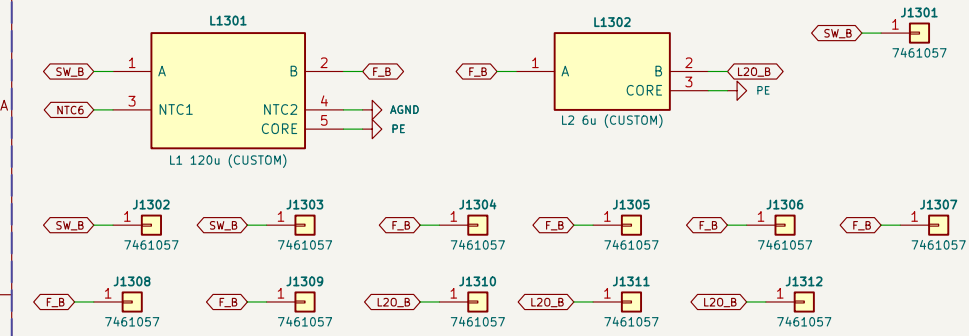
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 13
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /12 Phase a LCL filter, current sensor/
File: 12_filta.kicad_sch

Title: PCS-P125 power – Phase a LCL filter, current sensor

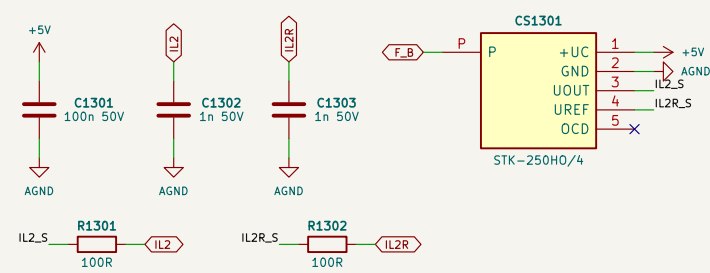
Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 13/28

PHASE B: L1 AND L2 (CHASSIS, CUSTOM) AND THEIR STUDS



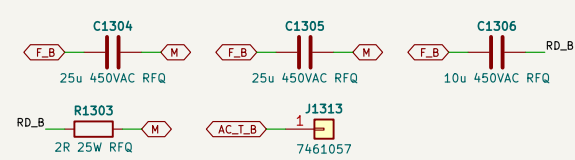
Board studs (REDCUBE press-fit, 100 A each): 3 x switch node to L1, 3 x L1 out (C_f node, its busbar through the sensor), 3 x C_f node to L2, 3 x L2 out (to AC contactor K1 on the port sheet). Cores bonded to PE.

PHASE B CURRENT SENSOR STK-250H0/4 (C_F SIDE OF L1)



IL2 = Uref + 3.2 mV/A x I (I + = L1 -> C_f; body arrow, verify at Incoming inspection). IL2R = its Uref pin; 100R + 1 nF each at the source (as PV-PWR). Primary = L1's output busbar through the aperture (a 50 Hz-smooth node, not the switch node). OCD (open drain, 2.93 x I_PN = 732 A +/-10 %) not used: above both IL windows and the L1 limit.

PHASE B C_F AND DAMPING BRANCH (STAR = M, THE DC MIDPOINT)



C_f = 2 x 25 uF MKP (50 uF); R_d 2 ohm + C_d 10 uF in series, F_B to M.

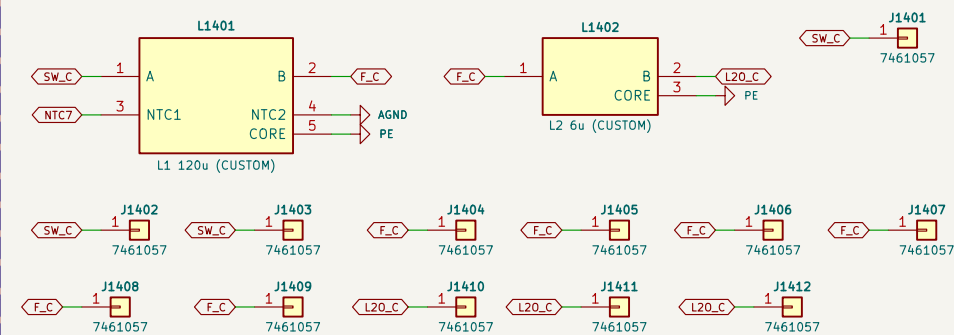
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 14
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /13 Phase b LCL filter, current sensor/
File: 13_filtb.kicad_sch

Title: PCS-P125 power – Phase b LCL filter, current sensor

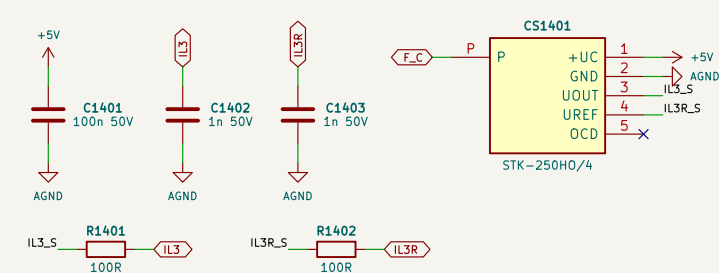
Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 14/28

PHASE C: L1 AND L2 (CHASSIS, CUSTOM) AND THEIR STUDS



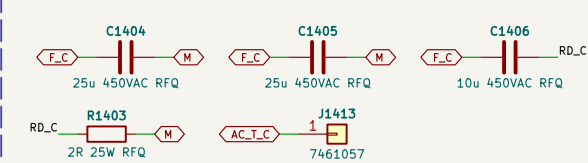
Board studs (REDCUBE press-fit, 100 A each): 3 x switch node to L1, 3 x L1 out (C_f node, its busbar through the sensor), 3 x C_f node to L2, 3 x L2 out (to AC contactor K1 on the port sheet). Cores bonded to PE.

PHASE C CURRENT SENSOR STK-250HO/4 (C_F SIDE OF L1)



IL3 = Uref + 3.2 mV/A x I (I = L1 -> C_f: body arrow, verify at incoming inspection), IL3R = Its Uref pin; 100R + 1 nF each at the source (as PV-PWR). Primary = L1's output busbar through the aperture (a 50 Hz-smooth node, not the switch node). OCD (open drain, 2.93 x L_{PN} = 732 A +/-10 %) not used: above both IL windows and the L1 limit.

PHASE C C_F AND DAMPING BRANCH (STAR = M, THE DC MIDPOINT)



C_f = 2 x 25 uF MKP (50 uF); R_d 2 ohm + C_d 10 uF in series, F_C to M.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 15
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /14 Phase c LCL filter, current sensor/
File: 14_filtc.kicad_sch

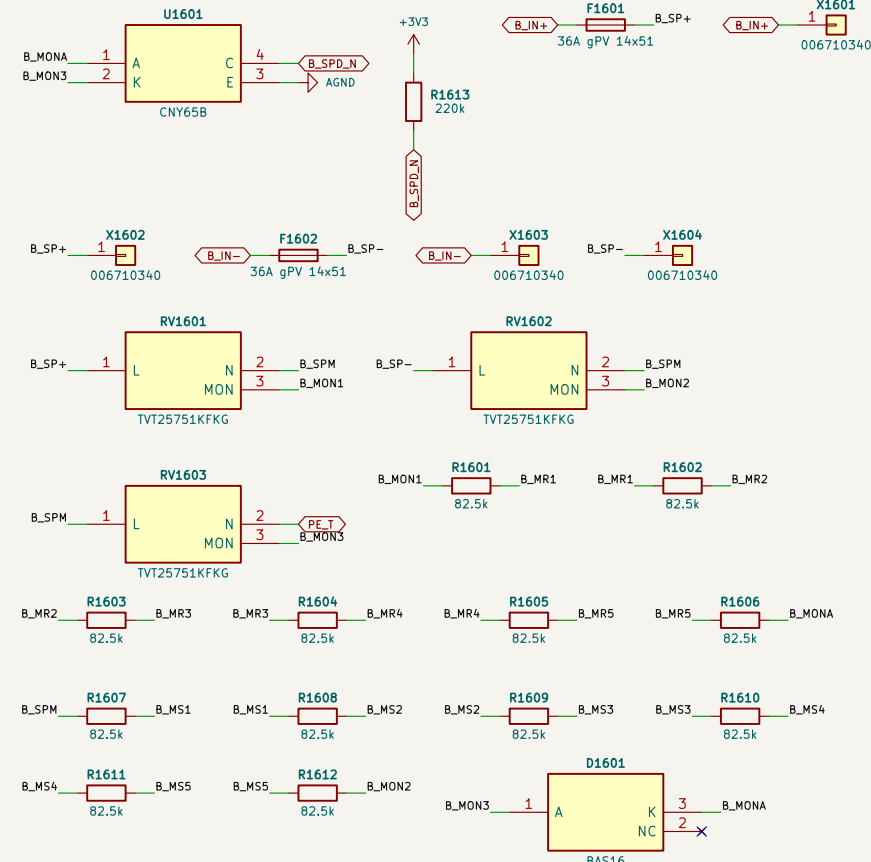
Title: PCS-P125 power – Phase c LCL filter, current sensor

Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 15/28

PORT B (BATTERY SOURCE), D-044 LEAN PORT

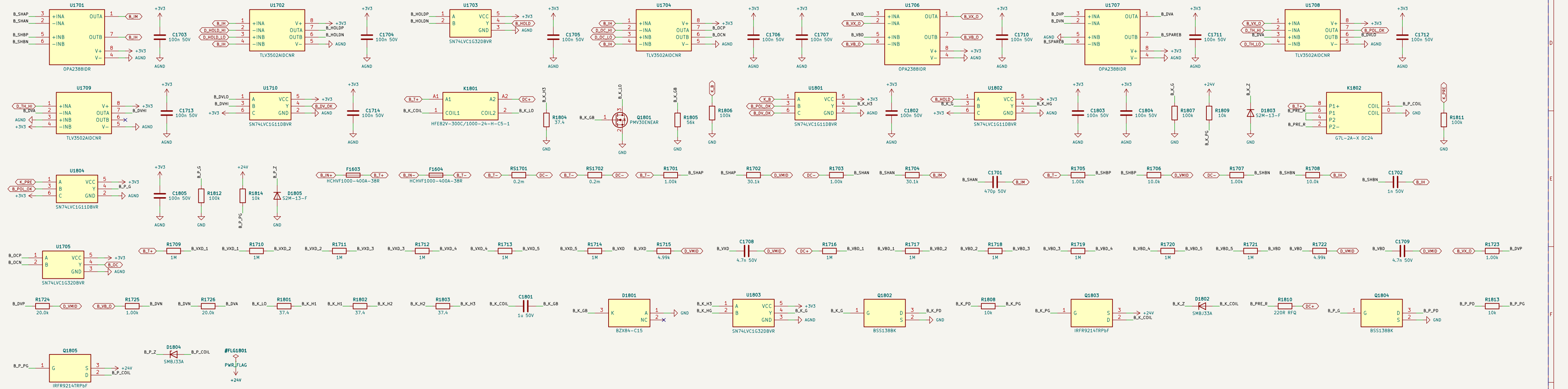
Terminal side B_{1N}+ / B_{1N}-, bank DC+. HCHVF1000-400A-38R aR in both poles.
Contactor opens <= 0.97 kA, held above (port_spec lean). Fuse / contactor coordination: the board's design check.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up,eff 3793 V pole-PE and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D=032); I_{max} 25 kA, Branch fuses ETI CH14x51 gpV36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

AR FUSES (BATTERY-TYPE PORT)



HCHVF1000-400A-38R in both poles (aR, partial range): see the board's design check.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_checks() Sheet 17 Schematics authored by Chinmoy Bhuyan Live board referenced to DC--; gate islands functional; HV-PE basic		
Sheet: /16 Port B power path (battery)/ File: 16_leanB_power.kicad_sch		
Title: PCS-P125 power – Port B power path (battery)		
Size: A2	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Date: 17/28

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 18
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /17 Port B sensing and interlocks/
File: 17_leanB_ctrl.kicad_sch

Title: **PCS-P125 power – Port B sensing and interlocks**

Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6	Id: 18/28	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 19
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /18 Port B coil drives/
File: 18_leanB_drive.kicad_sch

Title: PCS-P125 power – Port B coil drives

Size: A3

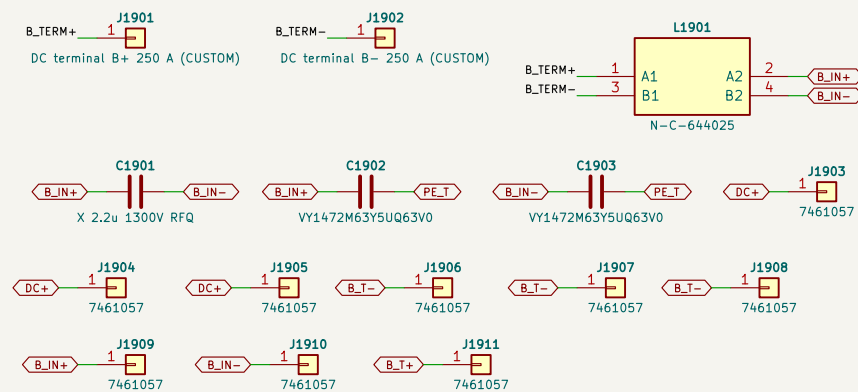
Date: 2026-10-06

Rev: A0

KiCad E.D.A. 10.0.6

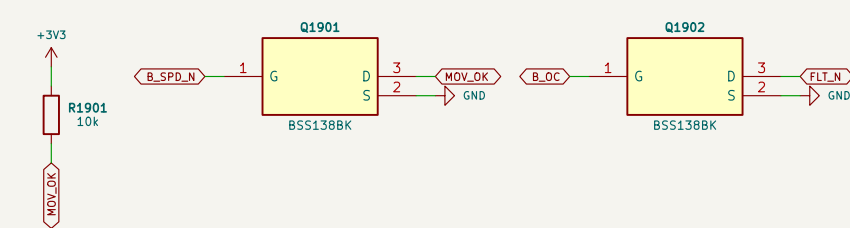
Id: 19/28

DC TERMINALS AND FILTER



B_TERM+/- (250 A), X 2.2 uF 1300 V across the port, Y1 4.7 nF pole - PE_T (removable PE link for the hipot).
Board taps B_IN+ / B_IN- by short links.

PORT STATUS TO PCS_PC



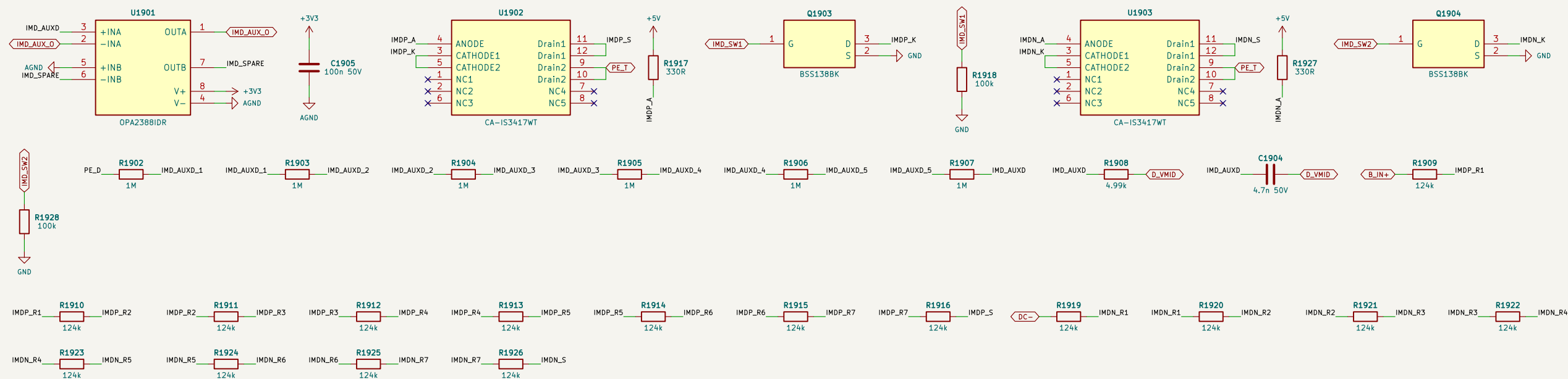
HOLD = B_HOLD (||| >= 0.97-1.03 kA: contactor held). MOV_OK high while the varistor monitor loop conducts. Port over-current (387-413 A) pulls FLT_N low (open drain): the latch trips without firmware.

PE



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot).

INSULATION MONITOR (PORT.LEAN_IMD)



Battery side, contactors open (IEC 62109-2 style check before connecting):
IMD_SW1 = string B_IN+ -> PE, IMD_SW2 = PE -> DC-; VPE to PCS_PC.

Not bench-validated. Values read from `sim/out/pcs_design/pcs_spec.json`; checks: `design_check()`
Sheet 20

Schematics authored by Chinmoy Bhuyan

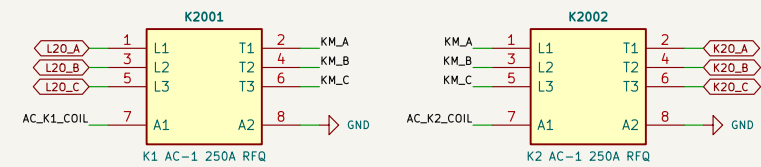
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /19 DC terminals, insulation test/
File: 19_dcterm.kicad_sch

Title: PCS-P125 power – DC terminals, insulation test

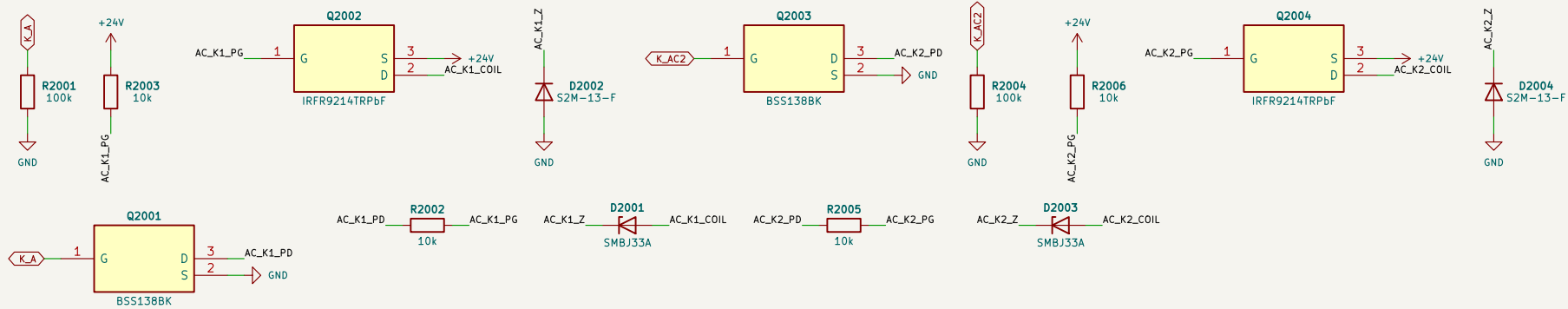
Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 20/28

AC CONTACTORS K1, K2 (CHASSIS, RFQ): TWO 3-POLE CONTACTORS IN SERIES



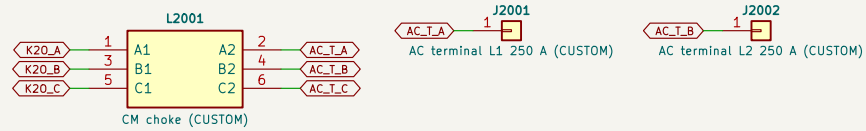
Relay test before every connection: K1 alone, then K2 alone; the voltage across the open one is $V_G - V_C$ (stage-3 dividers). Close only after synchronisation, open only at zero current (firmware).

COIL DRIVERS (PORT_LEAN_DRV): BSS138 -> IRFR9214 FROM THE LIVE 24 V, S2M + SMBJ33A CLAMP



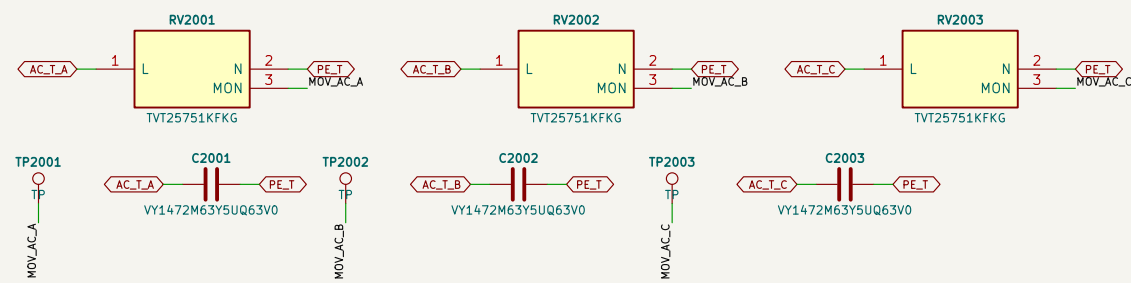
K1 from K_A (= K_AC1), K2 from K_AC2; logic lost = coil off. The contactors' DC coil modules economise.

CM CHoke (CHASSIS, CUSTOM) AND AC TERMINALS



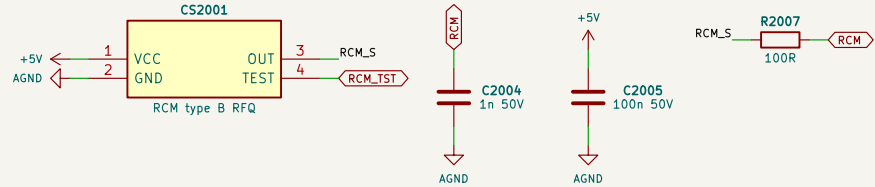
One turn of the three phase busbars through the cores; terminals L1-L3 250 A; PE terminal on the chassis.

AC SURGE PROTECTION (TYPE II) AND Y1



Per phase a thermally protected varistor TVT25751 (465 V AC) and a Y1 4.7 nF to PE_T; the removable PE link opens them for the hipot. Monitor leads to test points (OPEN 5).

RESIDUAL-CURRENT SENSOR (TYPE B, RFQ)



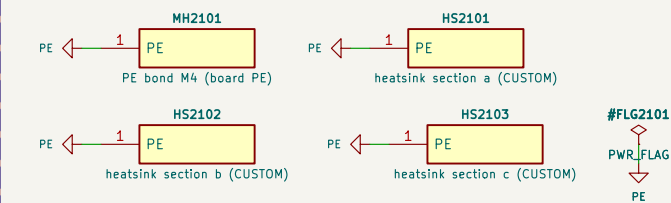
Around L1-L3 between the CM choke and the terminals; RCM analog out and RCM_TST to PCS_X.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 21
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

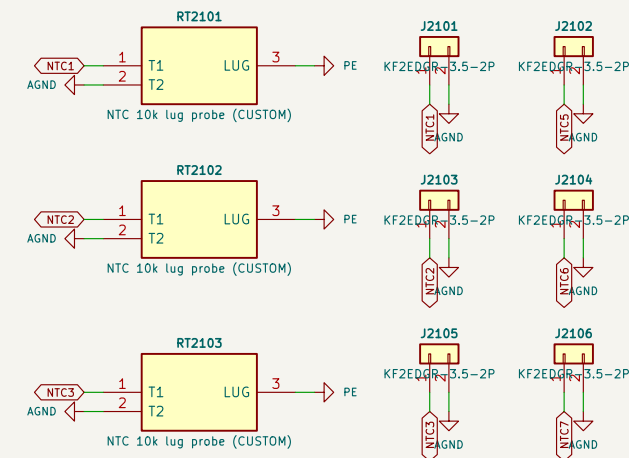
Sheet: /20 AC port: contactors, SPD, RCM/
File: 20_acport.kicad_sch

Title: PCS-P125 power - AC port: contactors, SPD, RCM

Size: A3 Date: 2026-10-06 Rev: A0
KiCad E.D.A. 10.0.6 Id: 21/28



The diagram illustrates a test setup with six parallel channels. Each channel contains a thermistor (labeled NTC1 to NTC6) connected to a capacitor (labeled C2101 to C2106). The capacitors are specified as 100nF 50V. The bottom terminal of each capacitor is connected to a common ground labeled AGND.



NTC1-3 = heatsink sections a-c: lug probe on the EARTHED heatsink, ≥ 2.5 kV rms lead-to-lug (R-10).
NTC5-7 = L1 winding NTCs (leads on 2-pin headers), 100 nF at each header.

24 V → +5V (LMR38020, 400 KHZ, 2 A)

U2201: LMR38020FDDAR

Inputs: VIN (24V), EN (RT_SYNC), RT_SYNC (RT_5V), FB (PG_5V), GND (GND), EP (GND)

Outputs: SW_5V, BOOT_5V, FB_5V, PG_5V, GND

Components:

- C2201: 4.7uF 50V
- C2202: 4.7uF 50V
- C2203: 100nF 100V
- R2201: 64.9k
- L2201: 74437368150
- C2205: 22uF 16V
- C2206: 22uF 16V
- C2207: 22uF 16V
- R2202: 100k
- R2203: 24.9k
- C2204: 100nF 50V
- R2204: 100k
- TP2201: TP

NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1–2, ctrl_c2000 circuit (SNVSC40E table 9–1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % → 4.93–5.10 V. Load in design_check (<= 80 % of 2 A).

[illegible]

gate bias of legs b and c, +3V3 (TP562130), ctrlLc2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP

The schematic shows the U2204 (TPS62130RQTR) regulator circuit. The input is +5V_GD (pin 11, 12, 10). The output is +3V3 (pin 14, 5). The circuit includes a feedback network with resistors R2211 (31.6k) and R2212 (10k), and a compensation network with capacitors C2217 (10uF 25V), C2218 (100nF 50V), C2219 (470pF 50V), C2220 (22uF 10V), and C2221 (22uF 10V). The chip is labeled U2204 and TPS62130RQTR.

EN = PG_5V6, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (≥ 3.20 V for the control board's reference). SS/TR 470p COG;
ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control ≤ 157 mA).

[illegible]

SELV 24 V TO THE CONTROL BOARD (J2)

Diagram illustrating the wiring for SELV 24 V to the control board (J2):

- SELV_24V (Green wire) is connected to pin 1 of the J2201 connector.
- SELV_0V (Red wire) is connected to pin 2 of the J2201 connector.
- The connector is identified as KF2EDGR-3.5-2P.

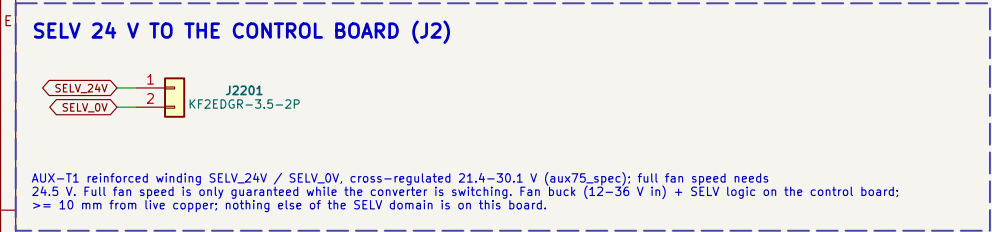
AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 21.4–30.1 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

SELV 24 V TO THE CONTROL BOARD (J2)

Diagram illustrating the wiring for SELV 24 V to the control board (J2):

- SELV_24V (Green wire) is connected to Pin 1 of the J2201 connector.
- SELV_0V (Red wire) is connected to Pin 2 of the J2201 connector.
- The connector is identified as KF2EDGR-3.5-2P.

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+5V_GD SUPERVISOR ON RDY

The diagram illustrates the circuit for the +5V_GD supervisor on the RDY pin. The supervisor IC, U2203 (TPS3710DDCR), is configured with its VDD pin connected to +5V_GD and its SENSE pin connected to SUP5G_SNS. The OUT pin is connected to RDY. The GND pins are connected to GND. A pull-up resistor R2209 (107k) is connected between +5V_GD and SUP5G_SNS. The RDY pin is connected to GND through a pull-down network consisting of a capacitor C2215 (100nF, 50V) and a resistor R2210 (10k) in parallel. A capacitor C2216 (1nF, 50V) is connected between SUP5G_SNS and GND.

+5V_GD SUPERVISOR ON RDY

The diagram illustrates the circuit for the +5V_GD supervisor on the RDY pin. The supervisor is a TPS3710DDCR (U2203). The VDD pin is connected to +5V_GD, and the SENSE pin is connected to SUP5G_SNS. The OUT pin is connected to RDY. The GND pins are connected to GND. A pull-up resistor R2209 (107k) is connected between +5V_GD and SUP5G_SNS. The RDY pin is connected to a pull-up resistor R2210 (10k) and a capacitor C2215 (100nF 50V). The SUP5G_SNS pin is connected to a capacitor C2216 (1nF 50V).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)

The diagrams illustrate the discharge of the +3V3 and +24V rails when the rails fail. The +3V3 rail is discharged through a 1.2R resistor (R2213) to GND. The +24V rail is discharged through a 100k resistor (R2214) to GND. The PG_5V0 rail is discharged through a 1.2R resistor (R2201) to GND. The BZ84-B10 diode is also shown.

PG_5V0 low (+5V_{DD} < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R; +3V3 falls at ≥ 20 mV/ μ s through the BOR region (F28003x SRVDDIO-DN).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)

The diagrams illustrate the discharge of the +3V3 and +24V rails when the rails fail. The +3V3 rail is discharged by a PMV30ENAR MOSFET (Q2201) and a BZX84-B10 diode (D2201). The +24V rail is discharged by a 100k resistor (R2214).

+3V3 Discharge:

- The +3V3 rail is connected to the drain of the MOSFET Q2201 (PMV30ENAR).
- The gate of Q2201 is connected to DIS_G (pin 1).
- The source of Q2201 is connected to GND.
- The diode D2201 (BZX84-B10) is connected between DIS_G and GND, with the cathode to DIS_G.

+24V Discharge:

- The +24V rail is connected to the drain of the MOSFET Q2202 (2N7002BK).
- The gate of Q2202 is connected to PG_5V_G (pin 1).
- The source of Q2202 is connected to GND.
- The resistor R2214 (100k) is connected between the +24V rail and GND.

PG_5V_G Low:

- When PG_5V_G is low, the MOSFET Q2202 is turned off, and the +24V rail is discharged through the resistor R2214.

TPS62130 Discharge:

- The MOSFET Q2201 (PMV30ENAR) is used to discharge the +3V3 rail when the rails fail.

TAP B_T+



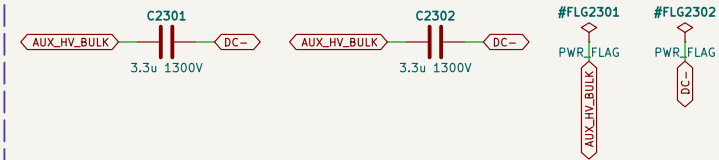
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP DC+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

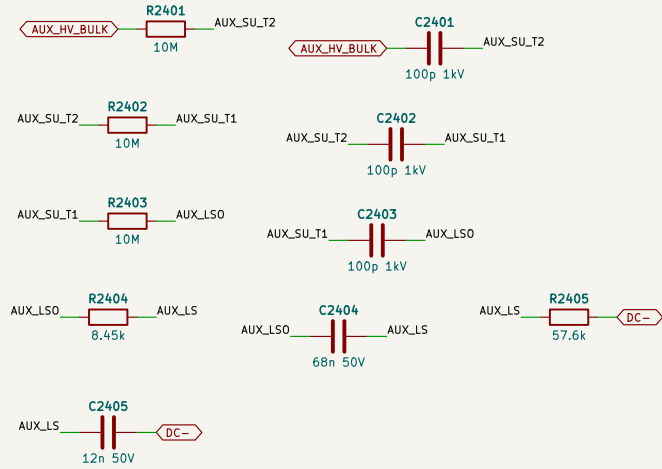
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 24
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /23 AUX75 input taps, OR, film/
File: 23_aux75_input.kicad_sch

Title: PCS-P125 power – AUX75 input taps, OR, film

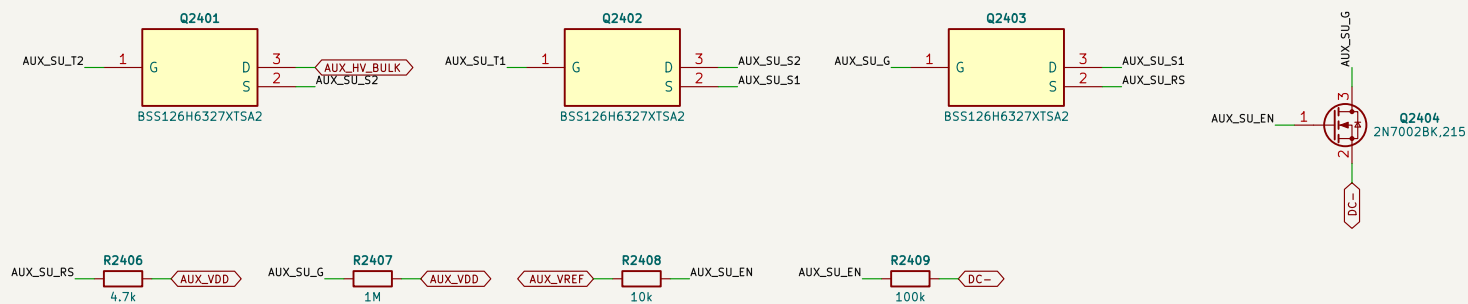
Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 24/28

COMPENSATED TAP STRING



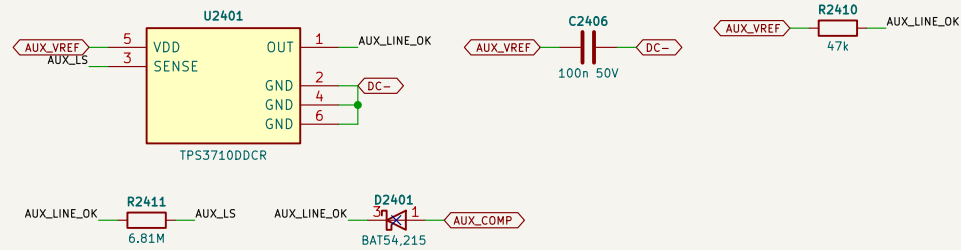
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



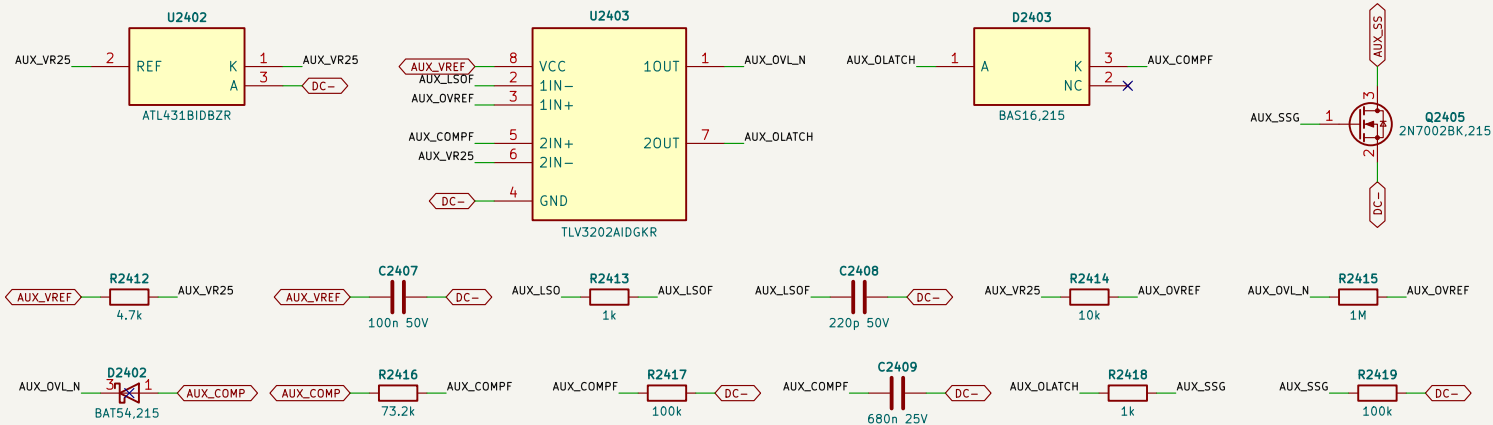
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 113 / 175 / 278 W (no trip at the 112.2 W 1 s peak); timer 39–100 ms -> latch -> SS crowbar -> hiccup.

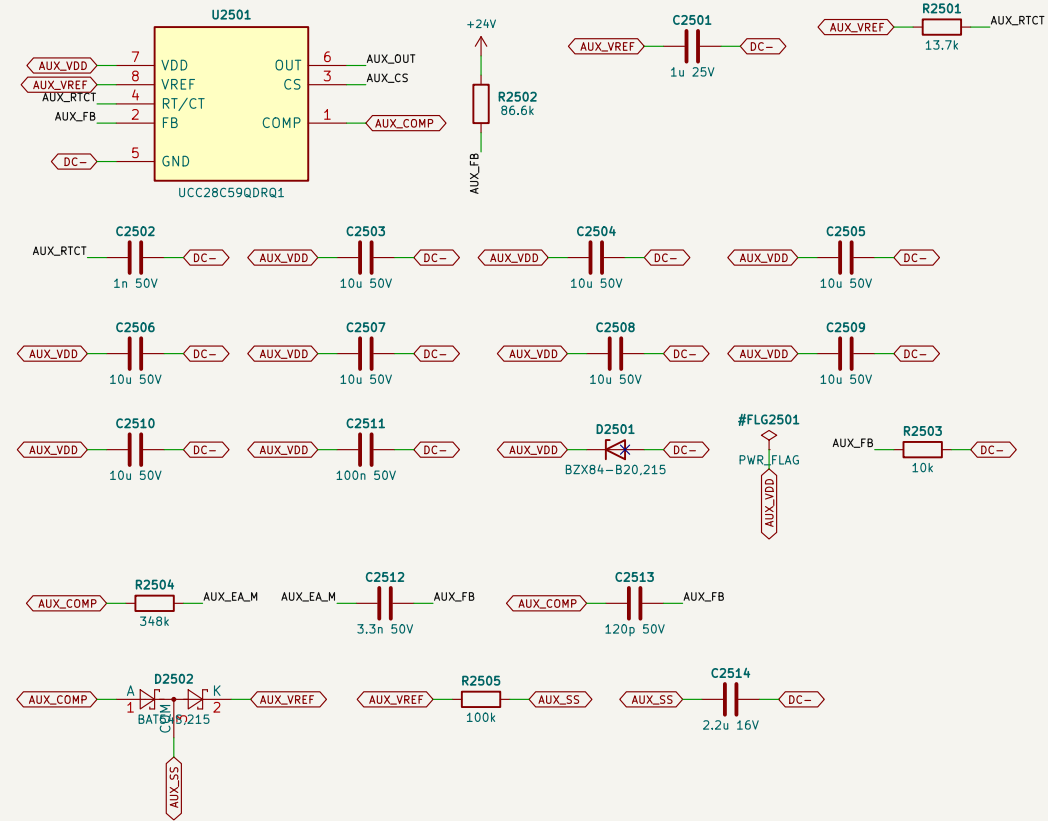
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 25
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /24 AUX75 start-up, brown-in, OV, timer/
File: 24_aux75_startup.kicad_sch

Title: PCS-P125 power – AUX75 start-up, brown-in, OV, timer

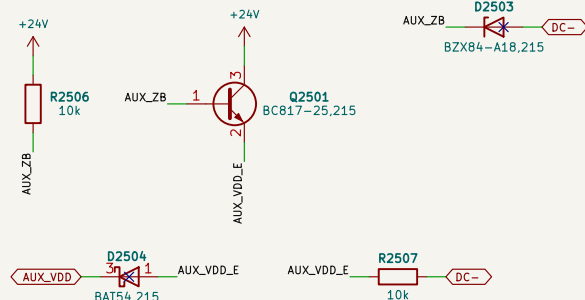
Size: A3 Date: 2026-10-06 Rev: A0
KiCad E.D.A. 10.0.6 Id: 25/28

PWM CONTROLLER AND ERROR AMPLIFIER



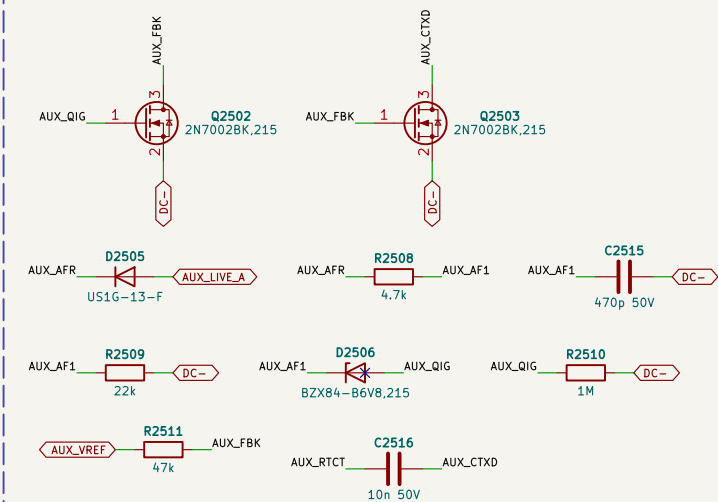
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 259–727 Hz, phase margin ≥ 60 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



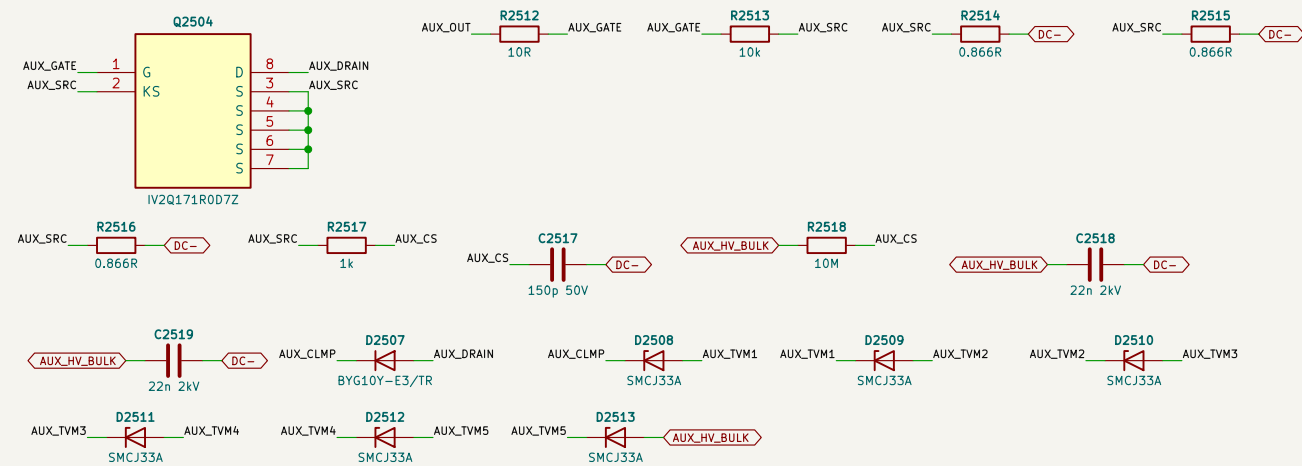
(30 W case, unchanged) follower Zener bias: VDD = gate drive 15.46–18.11 V (BZX84–A18, –40..85 °C)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



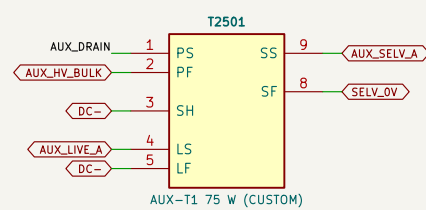
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



3 In parallel = 0.288667R (E96): lowest limit 3.04 A delivers 112.8 W (>= the 112.2 W 1 s peak) at every input; highest 4.07 A with the line feed-forward (4.46 A without) line feed-forward HV_BULK -> CS (CRH2512): CS offset 101 mV at 1000 V cancels most of the 155-250 nA CSJ44 overshoot, so the highest current limit no longer rises with the input Clamp: BYG10Y + 6 x SMCJ3A_VDS 1318 / 1422 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE’S REINFORCED SUPPLY BARRIER B1)



Lp 0.485 mH +/-7 %; shield and live winding on DC-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

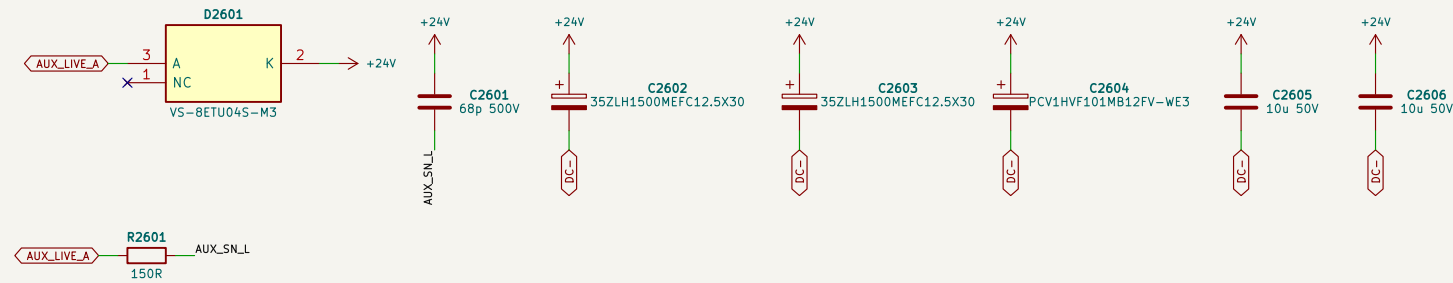
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 26
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /25 AUX75 PWM, SiC, clamp, AUX-T1/
File: 25_aux75_power.kicad_sch

Title: PCS-P125 power – AUX75 PWM, SiC, clamp, AUX-T1

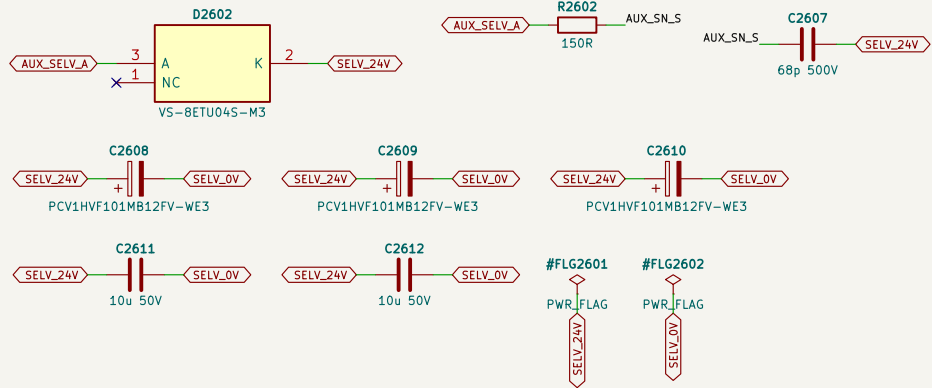
Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 26/28

LIVE 24 V RECTIFIER, HOLD-UP



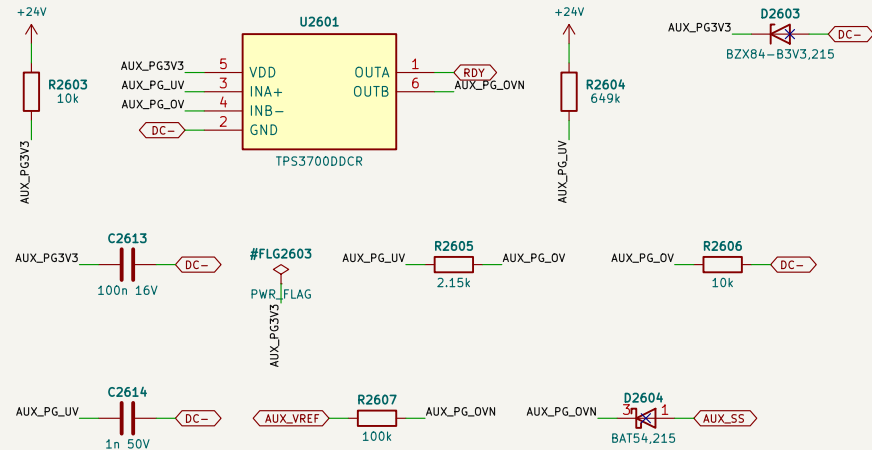
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)

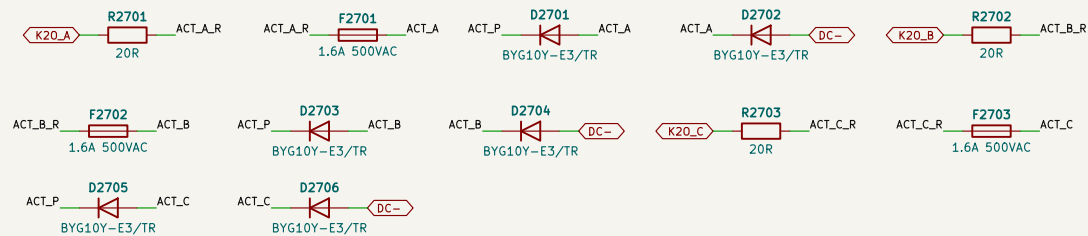
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 27
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /26 AUX75 live 24 V, SELV 24 V, status/
File: 26_aux75_outputs.kicad_sch

Title: PCS-P125 power – AUX75 live 24 V, SELV 24 V, status

Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 27/28

GRID TAP (LIVE DOMAIN: THE LOWER DIODES TIE DC- TO THE GRID PHASES; BASIC OVC III TO PE AS THE AC SIDE)



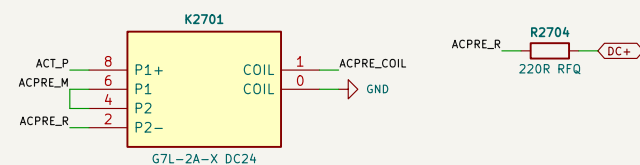
Per phase from K20_x (grid side of K1 / K2, works with the contactors open; behind the CM choke and the AC SPD):
AC10 20 ohm (inrush, surge and fault limit ahead of the fuse) -> ABB-A 1.60 (1.6 A, 500 V AC, 200 A interrupting) -> bridge.

OR INTO THE AUX INPUT (75 W BLOCK'S AUX_HV_BULK)



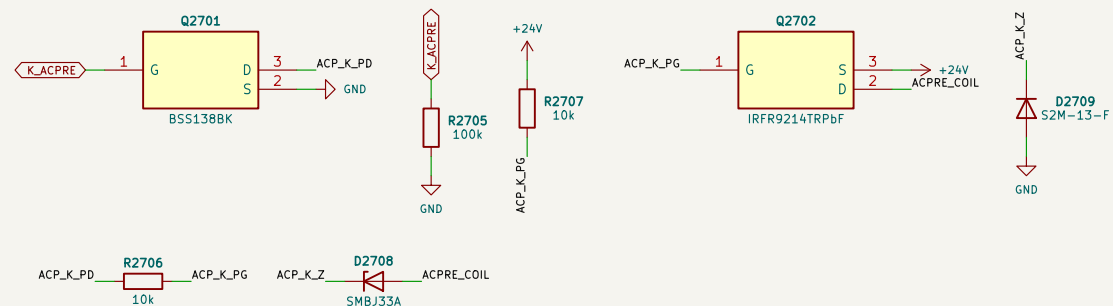
The tap's + rail feeds the aux through its own BYG10Y; the aux runs from the grid with the DC side dead (565-650 V rectified at 400-460 V AC, inside its 250 V..1000 V full-power range).

AC PRECHARGE: G7L-2A-X (DC, POLES IN SERIES, NORMAL POLARITY) + 220 OHM RPRE_AL FROM ACT_P TO DC+



Charges the DC link to the line-to-line peak through the tap with the main contactors open (pcs_spec ac_start);
firmware interlocks: K_ACPRE only with K_B, K_PRE, K_A, K_AC2 open; abort / retry / welded-relay rules.

COIL DRIVER (PORT._LEAN_DRV) FROM K_ACPRE



K_ACPRE = PCS_PC PWM16, gated by the trip latch on the control board; logic lost = coil off.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 28
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /27 AC start-up: grid tap, AC precharge/
File: 27_acstart.kicad_sch

Title: PCS-P125 power – AC start-up: grid tap, AC precharge

Size: A3	Date: 2026-10-06	Rev: A0
KiCad E.D.A. 10.0.6		Id: 28/28